

FIELD TECHNOLOGY V5

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Sovereign Time Sync — Terror-Threat Posture

Introduction — why sovereign time exists

Chapter 19 is the operator perimeter for **time**. Not cosmological time. Not poetic time. The clocks your logs, your GPS sub-micron nodes, your thermo receipts, and your threat panel agree on — or refuse to agree on, loudly, with a verdict you can `grep`.

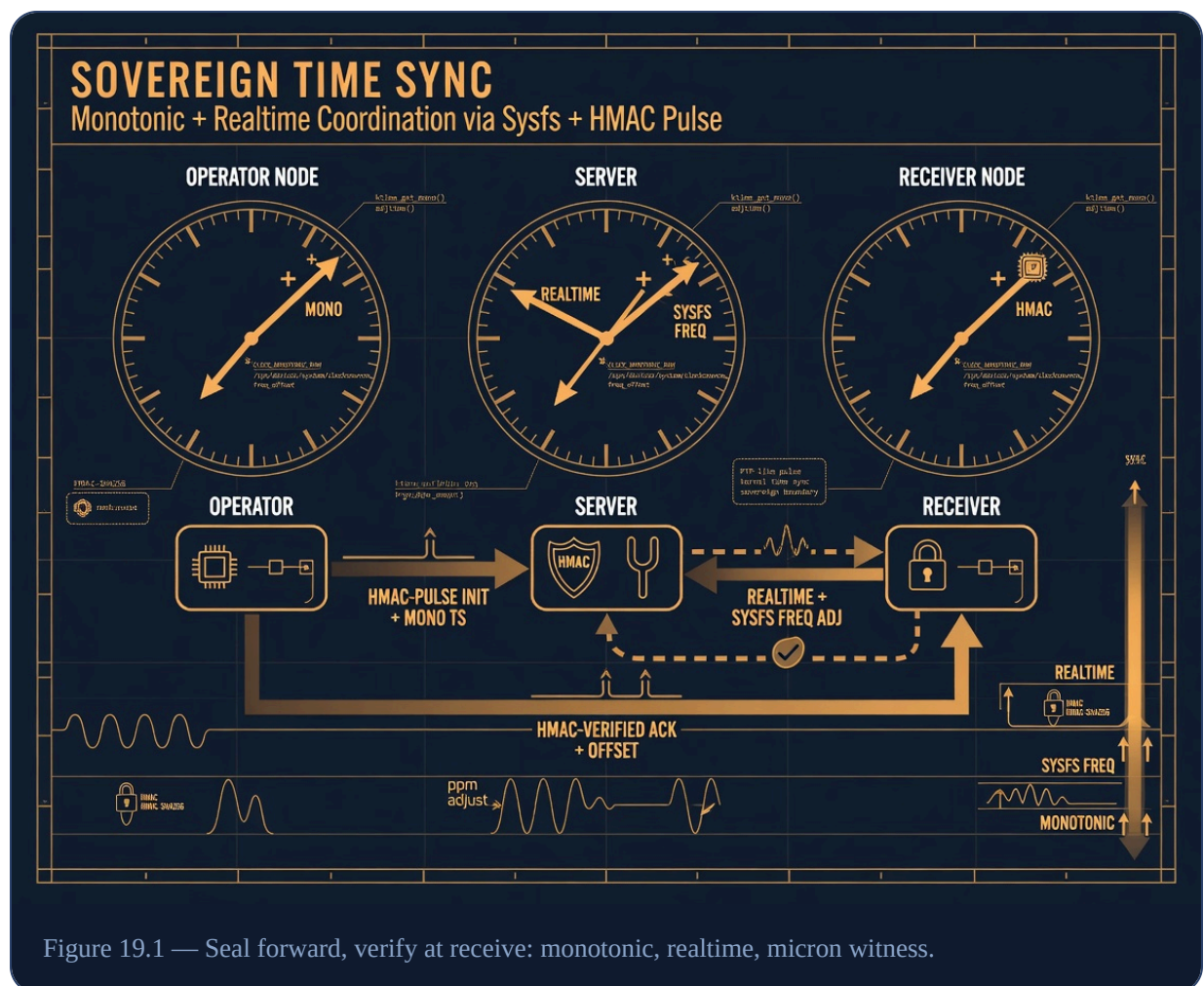
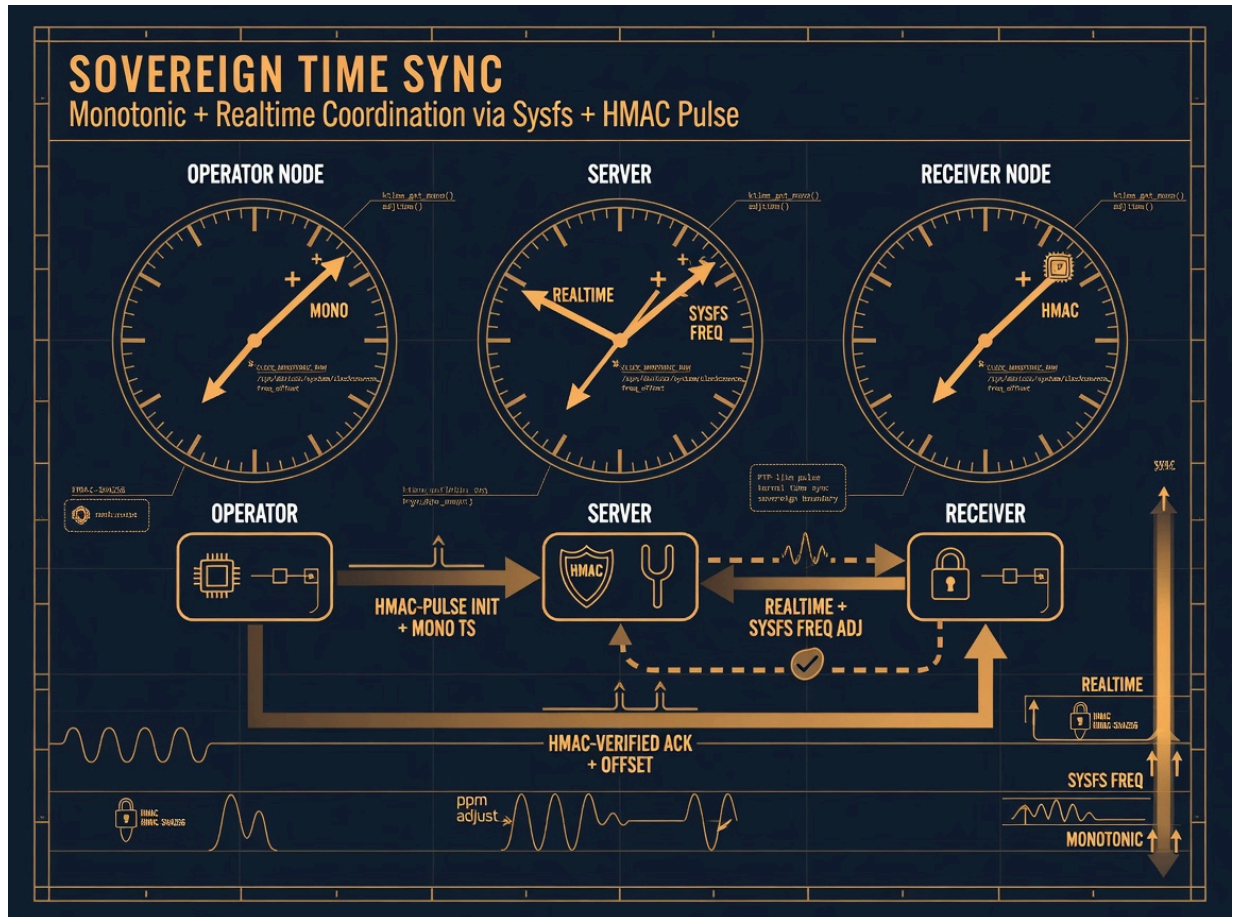


Figure 19.1 — Seal forward, verify at receive: monotonic, realtime, micron witness.

On the way — what you will learn

Sovereign time is operator-owned pulses — seal forward, verify at receive, grep SQUIDGIE when clocks disagree. On the way you will run `sovereign-time.py`, bind `monotonic/realtime/sysfs` witness, and fail closed under terror-threat posture.



On the way — Sovereign time

- Three clocks, three jobs
- HMAC receipts and micron_witness
- SQUIDGIE verdict catalog
- Extension of `TotalTime::seal()` across hosts

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Assume adversaries know field literacy. Assume they have read this book, or something like it. Assume pool NTP, remote RTC chips, hypervisor clocks, and silicon frequency tables can be **squidgied**: nudged just enough to desync correlated evidence without tripping naive alarms. A squidgie is not a cartoon hack. It is a micron-scale tamper — RTC nudged, monotonic story disagrees, or the sysfs CPU frequency fingerprint jumps without a logged thermal step that explains the jump.

Under terror-threat posture, you do not let the internet set your epoch alone. You run **your** timeserver. You sign pulses. You verify at receive. You seal forward. Entropy cannot be reversed; neither can honest time. This chapter teaches the mechanism, the grep discipline, and the honest rocks.

If you completed Chapter 11's observability lab, you already read STATUS lines as a timeline. Sovereign time makes that timeline *defensible* when adversaries target correlation rather than availability. Availability attacks are loud. Squidgie attacks are quiet. Quiet attacks are why verify-at-receive exists.

Threat model — what adversaries actually do to clocks

Pool NTP is a convenience mesh, not a covenant. Clients ask strangers for the time; strangers answer; clients step their clocks. Under terror-threat knowledge, that model leaks trust at every hop. An adversary who controls any influential stratum source — or merely influences RTT — can nudge many hosts together, hiding tamper inside statistically normal offset noise.

Remote RTC tamper is subtler still. Battery-backed clocks on motherboards and BMCs survive reboots. A small persistent bias accumulates across days until log correlation between your engine thermo receipts and your NEXUS packet field sentences drifts. Forensics then mis-orders cause and effect: did the connection precede the entropy spike, or did bad wall labels make it look that way?

Hypervisors add a third surface. Guest CLOCK_REALTIME can be warped by host policy without the guest's monotonic story noticing immediately. That is why monotonic alone is insufficient for cross-host agreement: monotonic is session-local truth, not universal UTC truth.

Sysfs frequency tables are the Puny-tier witness Spiderweb already reads. Adversaries with kernel foothold can spoof scaling_cur_freq reads while thermal governor narratives lag. Micron witness binds monotonic instants to those tables so fast flips without DVFS story become first-class issues, not debugging curiosities.

Field Technology does not claim magic detection of every FPGA timestamp attack. It claims **Implemented** operator-owned pulses, signed receipts, and receiver double-check with a named

realtime labels on enrolled hosts trace to **your operator timeserver pulses**, not pool consensus alone.

The **sysfs freq witness** reads per-core `scaling_cur_freq` (or `cpuinfo_cur_freq`) at pulse instants. Spiderweb Puny tier already treats these MHz tables as honest silicon telemetry for dashboards. Sovereign time hashes them into `micron_witness` so pulse receipts carry a silicon fingerprint, not only software clocks.

Naive NTP clients collapse these questions into one adjusted offset. Field Technology refuses that collapse. When GPS precision nodes report sub-micron ENU coordinates but sovereign verify returns SQUIDGIE, trust the triple verify — not the prettiest map dot.

Session-local spine — `TotalTime::seal()`

Inside AMOURANTHRTX, `TotalTime::seal()` in `ELLIE.hpp` locks session genesis into `FieldSocket::sealed_time`. Frame-rate jitter cannot rewrite physics time. That is **session-local** monotonic discipline: one engine process, one sealed genesis, one forward-only story per run.

Sovereign sync extends the same posture *across hosts*. The engine seals time inside the dispatch loop; the operator node issues signed pulses on UDP; every receiver pulls a pulse and runs `verify_receipt()` before accepting wall labels for perimeter services. Captain Ellie seals time forward in C++; `sovereign-time.py` seals it forward on the LAN mesh. Same ethics, different scale.

The coupling is intentional. Operators who run Queen browser beside AMOURANTHRTX should not maintain two incompatible time stories. Sealed session time governs fabric evolution; sovereign pulses govern perimeter services and log correlation across processes. Chapter 21 binds Queen navigation receipts to the same witness Chapter 19 verifies.

When `TotalTime::verify()` fails in-engine, abort posture is fail-closed. When sovereign verify fails, perimeter daemons gate replies — NTP blocks, panel flags red, grep finds SQUIDGIE. The C++ abort and Python gate are siblings, not duplicates.

SQUIDGIE — the tamper verdict

SQUIDGIE is the sovereign-time tamper verdict returned by `verify_receipt()` when issues are non-empty. It is not humor. It is the word you grep when clocks disagree at receive.



Figure 19.2 — SQUIDGIE verdicts belong in stderr timelines, not vibes.

Detection paths include `bad_signature` when HMAC from the operator key in `NEXUS_STATE_DIR` does not match the receipt body; `monotonic_backward` when remote `mono_ns` decreases versus the previous pulse; `realtime_backward` when remote `realtime_ns` decreases; `mono_real_skew` when realtime and monotonic deltas diverge beyond `NEXUS_TIME_MAX_SKEW_MS` (default 50 ms).

Hardware-story paths include `micron_squidgie` when `micron_witness` flips in under 50 ms of monotonic window while frequency sum jumps or skew is high; `freq_rtc_squidgie` when `freq_sum_khz` delta exceeds `NEXUS_TIME_MAX_FREQ_DELTA_KHZ` (default 250000) with suspicious skew. Receive-side `receive_wall_skew` compares local realtime to remote at pull time with generous RTT slack.

Verdict is either `USER_OK` or `SQUIDGIE`. There is no partial credit. Perimeter services treat `SQUIDGIE` as fail-closed input. `field-ntp-2026.py` increments `squidgie_blocks` when sovereign cache is dirty. Operators treat repeated squidgie as incident response, not calibration trivia.

The name evokes silent drift — squid ink in the water. Your job is to make ink visible in jsonl. Archive receipts. Correlate with thermo. Do not silence the verdict because the panel is pretty.

Micron witness — binding monotonic to silicon

Each pulse carries `micron_witness`: the first 16 hex characters of SHA-256 over `mono_ns` concatenated with sorted per-CPU frequency entries from `sysfs`. The witness binds a monotonic instant to the silicon table at that instant.

If adversaries nudge RTC silently, monotonic may still advance while realtime labels drift. If adversaries manipulate reported frequencies without thermal narrative, witness flips faster than DVFS alone explains. The witness is not electron microscopy. It is Implemented correlation discipline aligned with Spiderweb Puny tier reads.

Operators should sample witness across idle and load. Under honest DVFS, witness changes should correlate with workload steps you can see in Spiderweb status or `entropyThisFrame` under probe injection. Witness flips during apparent idle with simultaneous freq sum jumps are exactly the squidgie story.

Micron witness also supports enrollment audits: new hosts pulling pulses should show stable witness continuity when clocks are honest. A fresh host with wild witness divergence on first sync is a enrollment stop signal, not a prompt to widen NTP.

sovereign-time.py — anatomy and artifacts

The module lives at `NEXUS_INSTALL_ROOT/lib/sovereign-time.py`. Schema `sovereign-time/v1`. State defaults to `NEXUS_STATE_DIR (/var/lib/nexus-shield)`.

Figure 19.3 — HMAC receipts beside field jsonl — grep the verdict.

Claim: NEXUS packet field — local jsonl perimeter

Artifacts: `sovereign-time-key.bin` holds the 32-byte HMAC key (mode 0600 on create); `sovereign-time-pulse.json` stores pulse counter and last receipt; `sovereign-time-receipts.jsonl` is append-only audit. UDP port defaults to `NEXUS_SOVEREIGN_TIME_PORT=9123`.

Commands: `serve` runs operator UDP loop; `pulse` issues one local signed pulse; `sync [host] [port]` pulls and verifies; `status / json` emit panel slice. Verify mode accepts saved receipt JSON for offline forensics.

Each receipt includes schema, pulse number, chain label, host, mono_ns, realtime_ns, utc ISO, freq_sum_khz, micron_witness, entropy_tag (12 hex chars), and sig (HMAC-SHA256 over sorted JSON body). UDP server accepts {"op": "pulse", "client": "hostname"} and returns fresh signed receipt.

Default bind is loopback via NEXUS_SOVEREIGN_TIME_BIND. Widening to LAN is explicit operator choice — terror-threat posture assumes loopback until you document otherwise in your covenant file.

Receiver verify — pipeline step by step

Every receiving end that cares about honest wall labels runs sync on schedule or gates daemons on sovereign health.

Step one: pull signed receipt via UDP. Step two: load previous receipt from local pulse state. Step three: recompute HMAC with operator key material. Step four: compare monotonic and realtime deltas pulse-to-pulse. Step five: evaluate freq sum delta against MAX_FREQ_DELTA_KHZ. Step six: evaluate micron witness flip speed. Step seven: cross-check local clocks at receive. Step eight: write verdict and issues to pulse state; append to jsonl on issue.

Grep discipline: `grep SQUIDGIE /var/lib/nexus-shield/sovereign-time-receipts.jsonl` and inspect `sovereign-time-pulse.json` last verify block. Pair with threat-panel sovereign slice from `field-services-2026.py json`.

Offline verify: `python3 sovereign-time.py verify RECEIPT.json PREV.json` replays checks without network — useful when archiving incidents.

Do not auto-retry sync into silence. Repeated USER_OK after SQUIDGIE requires demonstrated fix: clock source corrected, key rotation documented, thermal story matches freq table.

Environment variables — operator reference

NEXUS_SOVEREIGN_TIME enables sovereign time in the NEXUS daemon stack.

NEXUS_SOVEREIGN_TIME_BIND defaults to 127.0.0.1 for serve mode.

NEXUS_SOVEREIGN_TIME_PORT defaults to 9123.

NEXUS_SOVEREIGN_TIME_FIRST=1 (default in 2026 seed) makes sovereign pulses authoritative over pool NTP — see `grok_world.sh` flow 20. NEXUS_TIME_MAX_SKEW_MS defaults to 50 ms. NEXUS_TIME_MAX_FREQ_DELTA_KHZ defaults to 250000.

NEXUS_STATE_DIR centralizes keys and receipts. NEXUS_INSTALL_ROOT locates lib modules for panel importers. Document these in your operator runbook beside DNS and DHCP binds.

Changing skew bounds is not tuning for comfort. Tighter bounds catch smaller squidgies; looser bounds reduce false positives on busy lab hosts. Production last-host nodes should justify any loosening in writing.

Coupling to precision GPS and Spiderweb

gps-precision.py sub-micron ENU nodes need stable UTC labels for map correlation. If sovereign verify fails while GPS draws confident dots, the dots are art until clocks recover.

Spiderweb Adept tier SimulateSubMicron uses sysfs clocks as freq witness alongside fabric averages. Sovereign micron witness should not wildly diverge from Puny-tier sysfs reads on the same host without explanation.

Cross-host GPS fusion assumes time agreement. Sovereign sync is the cheap test before trusting fused tracks. Skip it and you build pretty multi-host stories on sand.

ThermoAccountant entropyThisFrame is the per-frame receipt that time ran forward in fabric. Clock tamper does not directly spoof thermo — but mis-ordered logs make thermo spikes look like network causes. Fix time first; then grep thermo.

Coupling to NEXUS panel and field-ntp-2026

Threat panel merges sovereign slice from field-services-2026.py: posture flags, last pulse, last verify, squidgie blocks on NTP side.

field-ntp-2026.py serves RFC 5905 mode-4 on UDP 123, gated on sovereign pulse health. Stratum defaults to 2; NEXUS_LAST_HOST=1 can elevate stratum 1 on sole survivor node.

When sovereign cache is dirty, NTP replies may stop — clients should fail visibly rather than drift to pool silently. That is sovereign-first ethics.

Panel DNS tab time column should match grep-able receipts. If panel says OK but jsonl says SQUIDGIE, trust jsonl and file a panel stale-cache bug — not the other way around.

Single-source discipline — grok_world.sh flow 20

Gradient chaos from many NTP sources helps adversaries hide micron nudges. Flow 20 enforces single source when `NEXUS_SOVEREIGN_TIME_FIRST=1`: your signed pulses win.

This is not anti-NTP ideology. Stratum labels still matter — but must trace to verifiable pulses. Pool fallback remains opt-in via `NEXUS_POOL_NTP_FALLBACK=1`, never default in 2026 seed.

Operators document upstream when last-host mode serves global TIME. Sole survivor is not casual toggle — it is covenant posture (Chapter 20).

ELLIE alignment — seal forward across covenant

Captain Ellie's `TotalTime::verify()` abort posture is the spiritual parent of sovereign SQUIDGIE. Session entropy corruption triggers apocalypse handler fail-closed in C++.

`ellie-last-host.py verify` checks entropy covenant; `sovereign-time.py sync` checks clock covenant. Under `NEXUS_LAST_HOST=1`, both run on sole survivor — DNS, DHCP, TIME, grep one story.

`FieldSocket::sealed_time` in engine push constants parallels sovereign pulse → NTP stratum chain in perimeter. Teach both in onboarding.

Forensic workflow — week-two operator lab

Start operator timeserver: `python3 sovereign-time.py serve`. Issue baseline: `python3 sovereign-time.py pulse`. Sync from second context: `python3 sovereign-time.py sync 127.0.0.1 9123`. Confirm verdict: `USER_OK`.

Archive `sovereign-time-receipts.jsonl` with threat-panel exports. Correlate with `LOG_THERMO` during same window. Lab-only: induce VM skew; confirm SQUIDGIE and issues list populated.

Document recovery steps in runbook: stop serve, fix clock source, re-pulse, verify three consecutive OK, then re-enable NTP gate.

Troubleshooting common operator mistakes

Mistake: running serve on 0.0.0.0 without firewall story. Fix: bind loopback or LAN IP explicitly; document in covenant.

Mistake: copying key.bin between hosts without enrollment procedure. Fix: treat key as operator root; rotate on compromise.

Mistake: ignoring SQUIDGIE because browsing still works. Fix: browsing is not time truth; perimeter may be lying quietly.

Mistake: max skew set to seconds to silence alerts. Fix: you built a squidgie muffler; restore bounds or admit defeat in writing.

Honest rocks

UDP signed pulses on localhost: **Implemented** . HMAC verify + micron witness + freq fingerprint: **Implemented** .

Tamper abort like `TotalTime::verify()` in all NEXUS daemons: **Posture** — engine abort is C++. Sub-micron SEM from clock sync alone: **Correlation discipline** .

Pool NTP fully retired globally: **Operator policy** — sovereign-first is default, not universal law.

Case study — correlating thermo spikes with clock skew

An operator notices entropy spikes in LOG_THERMO coinciding with unexplained packet field bursts. Screenshots blame a browser tab. Sovereign grep shows SQUIDGIE entries 200 ms before the spike cluster — wall labels were wrong, not the fabric.

Recovery: stop NTP serve to clients, re-sync sovereign, confirm three USER_OK pulses, re-export jsonl with corrected timeline. The tab may still be guilty — but time truth comes first in forensic ordering.

Lesson: thermo is forward-running receipt; sovereign time is correlatable wall label. Neither replaces the other. Together they beat screenshots.

Case study — lab VM clock nudge

In a controlled lab VM, advance guest realtime by 400 ms without touching monotonic. Next sovereign sync returns `receive_wall_skew` and possibly `mono_real_skew` issues. Verdict SQUIDGIE.

Document issues list in incident notes. Revert VM clock. Re-pulse until `USER_OK`. This drill proves the stack works — run it before production enrollment.

Pulse chain integrity and `entropy_tag`

Each pulse increments a monotonic pulse counter stored in `sovereign-time-pulse.json`. Chain labels record whether pulse was `operator-local` or `serve:client` issued.

`entropy_tag` is 12 hex chars from SHA-256 over `mono_ns` and pulse number — lightweight tie between thermo ethic and time receipts. Grep `entropy_tag` beside `LOG_THERMO` hashes when building cross-subsystem timelines.

Key rotation and compromise posture

`sovereign-time-key.bin` is operator root for HMAC. Compromise requires rotate: stop serve, archive old jsonl, generate new key, re-enroll receivers, document rotation in covenant file.

Do not `chmod` key world-readable. Do not store key in git. Treat loss of key like loss of DNS admin passkey — full perimeter re-seal.

Integration checklist — before declaring production time

Confirm `NEXUS_SOVEREIGN_TIME_FIRST=1` in daemon env. Confirm serve bind matches network diagram. Confirm `field-ntp-2026` `squidgie_blocks` visible in panel. Confirm `chrony` disabled or hooked per policy.

Confirm Queen browser slice shows `sovereign_time_witness` true. Confirm `gps-precision` scripts read UTC from same receipt chain. Archive baseline receipts.jsonl.

Reading receipts.jsonl — field forensic grammar

Each jsonl row has ts ISO and receipt object. Sort by pulse number for chain analysis. Diff freq_sum_khz between consecutive pulses during CPU stress — expect gradual change, not step discontinuities without workload.

Pair micron_witness changes with Spiderweb STATUS lines. Mismatch between witness and Puny-tier sysfs manual read indicates deeper kernel spoof — escalate, do not tune skew.

Comparison to naive ntpdate behavior

ntpdate-style step changes wall clock instantly without signed chain or silicon witness. Under terror threat that is unacceptable for enrolled hosts. Sovereign sync is pull-verify-store, not blind step.

Clients may still use NTP for coarse sync if gated — but authority is sovereign pulse health, not pool vote.

Multi-host mesh — LAN enrollment story

Second host syncs to operator serve bind on LAN IP when covenant allows. RTT adds receive_wall_skew slack — defaults allow generous milliseconds on LAN, tighter on WAN if you must widen bind.

All enrolled hosts should share verification policy documentation — max skew, max freq delta, response playbook for SQUIDGIE.

Captain Ellie LOG categories and time

LOG_MAIN and LOG_STATUS timestamps should align with sovereign utc field after sync. If engine log and sovereign utc diverge systematically, suspect dual clock sources on one host — pool still running beside sovereign.

Disable pool when sovereign-first — grep pool in process list and config.

Spiderweb Puny tier — manual witness corroboration

Run list Hardware in prompt terminal while pulsing. Compare operationalFreqMHz narrative to freq_sum_khz in receipts. Large unexplained gaps trigger review even if verdict USER_OK — USER_OK is not absence of all anomaly, only absence of threshold issues.

ThermoAccountant and time — philosophical alignment

Landauer discipline says erase has cost. Time discipline says forward seal has receipt. ThermoAccountant counts proxy entropy per frame; sovereign-time counts proxy entropy_tag per pulse. Both refuse silent rewind.

When to widen NEXUS_SOVEREIGN_TIME_BIND

Widen only with documented LAN trust boundary — firewall rules, MAC allowlist, VPN. Loopback is default because UDP signed pulse is powerful; exposing serve to internet without story is retired-vulnerability thinking.

FAQ — will sovereign time fix my laptop sleeping

Sleep discontinuities affect monotonic versus realtime relationship across suspend — expect issues after resume until re-sync. Document resume playbook: pulse, sync, verify OK before serving NTP to others.

FAQ — is SQUIDGIE a false positive on laptops

Aggressive DVFS on battery can move freq_sum_khz quickly. Tune MAX_FREQ_DELTA only with logged thermal narrative — do not silence to comfort. Compare witness flip rate on AC versus battery in lab.

Long-form primer — sovereign time as operator covenant

Sovereign time is where Chapter 11 observability meets Chapter 18 covenant. You promised to build locally, grep truth, and hold gates. None of that works if wall labels lie. A threat panel timestamp that looks authoritative while RTC was squidgied is worse than no panel — it is false confidence. Sovereign time exists to deny false confidence a home on your perimeter.

Consider the full stack timeline on a single operator machine during a heavy session: AMOURANTHRTX dispatches fabric frames with `sealed_time` monotonic in `FieldSocket`; ThermoAccountant writes `entropyThisFrame` each tick; NEXUS captures socket sentences into `field jsonl`; Queen browser navigation adds honorability and WebRTC peer events; precision GPS may append sub-micron fixes; `sovereign-time.py` issues pulses tying realtime to `micron_witness`. Forensics after an incident require these streams to interleave without contradiction. If thermo spikes appear before the socket event that supposedly caused them because NTP stepped backward, you will chase the wrong process path. If GPS dots drift while clocks are clean, you suspect antenna or simulation tier — correct division of suspicion saves hours.

The operator timeserver is deliberately boring technology: UDP, JSON, HMAC, append-only `jsonl`. Boring is good. Fancy distributed consensus algorithms make pretty papers and ugly incident responses when you cannot explain who authorized a skew. Here the authority is the operator key in `NEXUS_STATE_DIR` and the `serve bind you` document. Pulses increment. Receipts append. Receivers verify. The story is grep-able start to finish.

Teach new operators three gestures before anything else: `serve`, `pulse`, `sync`. Then teach one grep: `SQUIDGIE`. Then open `field-services-2026` panel slice and point at `sovereign_time` block. If they understand those four surfaces, they will not panic when pool NTP disagrees — they already know which source wins under sovereign-first policy.

Micron witness deserves repetition because it is easily underestimated. Operators who live in software stacks forget silicon tables exist until performance tuning touches `cpufreq`. Spiderweb already surfaces MHz in Puny tier. Sovereign time hashes MHz into every pulse. That design choice links Chapter 10 hardware mirror literacy with Chapter 19 perimeter literacy — same `sysfs` read, two receipts. When witness flips fast, you are not required to immediately shout attacker; you are required to stop treating wall labels as evidence until the flip is explained. Explanation might be benign stress test; explanation might be squidgie. The verdict names the latter when thresholds trip.

Cross-host deployment adds latency but not complexity. Second machine pulls pulse, verifies signature with enrolled key material, compares deltas, writes local pulse state. `receive_wall_skew` accounts for RTT on LAN. Enrollment documentation should list allowed skew and forbidden clock sources on child hosts — `chrony` pointing at pool while parent serves sovereign is a common self-inflicted wound. `grep` processes and configs during onboarding.

Integration with field-ntp-2026 is the bridge to legacy clients that only speak NTP. Many tools still query UDP 123. Gating mode-4 replies on sovereign health means those tools fail closed when time is dirty — preferable to answering from a stepped pool clock. Stratum semantics remain meaningful: stratum 2 on normal nodes tells clients they are one hop from operator authority; stratum 1 on last host tells survivors there is no higher authority left on the wire. Do not abuse stratum 1 on everyday laptops — honesty labels matter in NTP sociology as much as in Chapter 12 rocks.

Captain Ellie alignment is not mythology. TotalTime::seal and TotalTime::verify are real C++ paths operators grep via engine logs. ellie-last-host.py verify is real Python path operators grep via covenant files. sovereign-time.py sync is real Python path operators grep via jsonl. Three seals, one ethic: forward only, verify before trust, abort or gate on failure. When students ask which seal is supreme, answer by scope: in-process fabric uses FieldSocket sealed_time; perimeter uses sovereign pulses; last host uses ellie-last-host seal at boot. Scopes nest, they do not compete.

Landauer and Shannon sit in the creditor chapters, but time is the hidden variable in every entropy story. Shannon surprise without timestamp is a bag of letters. Thermo entropy without timeline is a number without narrative. Sovereign time does not replace thermodynamic measurement — it orders measurement so narratives can be defended in front of another human who was not at your keyboard. That is operator covenant in practice.

Honest rocks close this primer: implemented UDP signed pulses, implemented verify paths, posture fail-closed parity with engine abort, metaphor correlation for sub-micron SEM claims, philosophy for pool retirement policy. We do not hide rocks. If your deployment cannot run serve on loopback because container networking forbids it, document the container exception in writing and widen bind with firewall story — do not pretend default loopback is optional because convenience complained.

Architecture diagram in prose — pulse lifecycle

Issue: operator host samples mono_ns, realtime_ns, sysfs freqs, computes micron_witness and freq_sum_khz, increments pulse counter, signs body, appends receipts.jsonl, optionally serves UDP response.

Transit: UDP datagram carries JSON receipt to enrolled receiver; no TLS in v1 — LAN trust boundary is operator responsibility; terror-threat posture assumes bind restriction and firewall.

Verify: receiver validates sig, compares to prev pulse, evaluates skew and witness flip rules, records verdict USER_OK or SQUIDGIE, updates local pulse state.

Consume: field-ntp-2026 reads sovereign cache; panel reads status json; Queen slice reads sovereign_time_witness flag; operator grep archives jsonl for incidents.

Recover: fix clock source, rotate key if compromised, re-pulse until three consecutive OK, re-enable gated services, document incident with issues list preserved — never delete SQUIDGIE rows from jsonl.

Operator runbook template — sovereign time incident

Detect: grep SQUIDGIE in receipts.jsonl or panel sovereign slice shows dirty cache. Contain: stop field-ntp-2026 client serve if you are downstream; do not delete jsonl. Diagnose: read issues list in last verify block — signature, skew, witness, freq. Remediate: eliminate dual clock sources; fix VM/host clock; rotate key if sig bad. Verify: three consecutive USER_OK from sync. Restore: re-enable NTP gate; archive incident bundle. Postmortem: correlate thermo and packet field with corrected timeline.

Clock	Role	Trust
Monotonic (CLOCK_MONOTONIC)	Ordering — ticks never go backward	Host kernel
Realtime (CLOCK_REALTIME)	Wall labels for grep and correlation	Your timeserver only
Sysfs freq witness	Spiderweb Puny tier — MHz per core	Read-only hardware

Seal forward. Session genesis and pulse chains only advance. Verify at receive. Every downstream host double-checks before trusting wall labels. Grep SQUIDGIE. Tamper is a verdict, not a vibe.

```
NEXUS_SOVEREIGN_TIME_BIND=127.0.0.1 python3 sovereign-time.py serve
python3 sovereign-time.py pulse
python3 sovereign-time.py sync 127.0.0.1 9123
grep SQUIDGIE /var/lib/nexus-shield/sovereign-time-receipts.jsonl
```

```
micron_witness = sha256(mono_ns | cpu0:freq | cpu1:freq | ...)[0:16]
TotalTime::seal() → FieldSocket::sealed_time (in-process)
sovereign-time.py pulse → HMAC receipt chain (cross-host)
```

Operator drills

Drill 19-A — Baseline seal. Serve, pulse, sync, archive receipts. Goal: three consecutive **USER_OK** verdicts with stable `micron_witness` across 60 seconds idle.

Drill 19-B — Grep discipline. Run engine + NEXUS 10 minutes. Grep SQUIDGIE in sovereign logs and THERMO in engine log. Goal: zero unexplained squidgie; thermo moves when fabric moves.

Drill 19-C — Freq witness read. While pulsing, read `scaling_cur_freq` manually. Goal: witness changes only on expected DVFS steps.

Drill 19-D — NTP gate. With `field-ntp-2026.py` running, confirm NTP replies stop when sovereign cache dirty after lab skew test.

Drill 19-E — Cross-host enrollment. Sync from second machine with shared state dir or enrolled key. Goal: document latency and `receive_wall_skew` within bounds.

Study questions

1. Why three clocks instead of `CLOCK_REALTIME` alone?
2. Difference between `TotalTime::seal()` and cross-host sovereign pulses?
3. Define SQUIDGIE using three detection paths from `verify_receipt()`.
4. How is `micron_witness` computed; what does fast flip imply?
5. When GPS and sovereign time disagree, which wins and why?
6. What does `NEXUS_SOVEREIGN_TIME_FIRST=1` change?
7. Where are signing keys stored; what file mode on create?
8. How does `field-ntp-2026.py` gate on sovereign health?
9. Why is single-source NTP discipline flow 20 not anti-NTP?
10. What forensic artifacts do you archive after a SQUIDGIE incident?

Chapter 20 — Public DNS, DHCP & Time →

Sovereign time receipts should be archived with the same discipline as packet field jsonl and thermo log exports. When an auditor asks prove this connection happened before that entropy spike, you interleave three files by pulse number and utc — not by screenshot collage.

Pool NTP clients on the same host as sovereign serve are a configuration smell. grep unit files and crontab for `ntpd` `chrony` `systemd-timesyncd`. Sovereign-first means those services are disabled or slaved to gated NTP — document the chosen pattern.

Hypervisor guests enrolled in sovereign mesh should disable host time sync override or verify after every migration. vMotion without re-sync is a classic squidgie source in enterprise labs.

Precision GPS operators should log sovereign pulse number alongside each ENU fix in local tooling. Correlation field makes sub-micron claims defensible in postmortems.

Teaching sovereign time to students: use lab VM skew first, production stories second. Students remember SQUIDGIE after they cause it safely themselves.

The `entropy_tag` on pulses is twelve hex characters — small enough to grep, large enough to disambiguate adjacent pulses in merged logs.

Frame-rate jitter in `AMOURANTHRTX` cannot rewrite `sealed_time` — that invariant is why session-local and cross-host seals are named differently. Do not merge the concepts when writing runbooks.

NEXUS daemon restarts should not rotate signing keys automatically — stability of key material is feature for enrolled receivers.

When `LAST_HOST` mode serves global TIME, sovereign serve bind may widen — document firewall rules alongside covenant invocation.

Micron witness flips during kernel cpufreq governor stress tests are expected — witness is witness, not verdict alone. Verdict combines witness speed with skew and freq sum delta.

Sovereign time receipts should be archived with the same discipline as packet field jsonl and thermo log exports. When an auditor asks prove this connection happened before that entropy spike, you interleave three files by pulse number and utc — not by screenshot collage.

Evidence anchor — grep and sources

Major claims in this chapter anchored for reproducibility. Implemented = grep today; Metaphor = intuition; Philosophy = discipline.

Claim	Statement	Label	Evidence
TotalTime::seal()	Session genesis lock	Implemented	FieldSocket sealed_time
Sovereign pulse	HMAC time forward	Implemented	sovereign-time.py
SQUIDGIE	Verify fail verdict	Implemented	Terror-threat posture
Pool consensus alone	NTP trust	Metaphor	Insufficient – operator-owned

seal → pulse → verify → USER_OK | SQUIDGIE

NEXUS_SOVEREIGN_TIME_BIND=127.0.0.1 python3 sovereign-time.py serve

Source paths

- NEXUS-Shield/sovereign-time.py
- Navigator/engine/TotalTime.cpp

Chapter summary — before you turn the page

Sovereign time: seal forward, verify at receive, SQUIDGIE when clocks lie. Operator-owned pulses, not pool consensus alone.

Public DNS, DHCP & Time — 2026

Introduction — public services under terror-threat posture

Chapter 20 is the 2026 Grok rewrite of public infrastructure on the operator perimeter: DNS, DHCP, and time. Assume terror-threat knowledge everywhere. Services must ship without old vulnerabilities — loopback-first, operator-owned, verify at receive.

On the way — what you will learn

Public services 2026 — loopback-first DNS, DHCP, and NTP without old vulnerabilities. On the way you will deploy Truth Resolver at `127.0.0.1`, trace-from-root discipline, and WAN exposure only with explicit operator consent.

On the way — Public services

- Truth DNS — no ANY floods
- Field DHCP on LAN
- Sovereign NTP with verify-at-receive
- Perimeter services as field extension

IN THIS CHAPTER

- | | |
|---|--|
| 1. Introduction — public services under terror-threat posture | 5. Field DHCP v3 — issue only, verify always |
| 2. Three services, one posture | 6. Sovereign NTP — field-ntp-2026.py |
| 3. Truth DNS — resolver ethics | 7. field-services-2026.py — unified panel manifest |
| 4. DNS admin portal — no hardcoded production keys | 8. ELLIE Last Host — sole survivor mode |
| | 9. Receive-verify chain across services |

10. Queen and FieldFox inheritance

11. Opt-in WAN exposure

12. Phase 2 — honest roadmap

**13. Week-three operator lab — full stack
bring-up**

14. Troubleshooting

These are not three unrelated daemons. They are one posture expressed on three ports. Truth DNS refuses foreign resolver shortcuts. Field DHCP issues leases only after conflict checks and option-50 validation. Sovereign NTP answers on UDP 123 only when sovereign pulses are clean. Queen browser and FieldFox inherit Truth DNS lock. KILROY field package inherits the same covenant at syscall boundary when deployed as sovereign field (Chapter 21).

WAN exposure requires `NEXUS_FIELD_SERVICES_PUBLIC=1` — explicit, never default. Default is operator LAN and loopback truth.

Three services, one posture

Truth DNS binds `127.0.0.1` and `::1` by default on port 53 UDP/TCP. Field DHCP binds LAN IP on port 67 UDP, not `0.0.0.0` unless public mode. Sovereign NTP serves stratum-2 (or stratum-1 on last host) from signed pulses on port 123.

Retired vulnerabilities are listed in `field-services-2026-seed.json`: hardcoded DNS admin passkeys, rogue DHCP on `0.0.0.0`, pool NTP as sole authority, dig fork bombs, option-50 mismatch, missing ping conflict detect, cleartext admin on `7/77/777` without localhost bind.

Unified panel slice: `python3 field-services-2026.py json` merges DNS, DHCP, NTP, sovereign time, and ellie-last-host posture.

Truth DNS — resolver ethics

Truth Resolver follows RFC 1034 `dig +trace` from root — no Google/Cloudflare shortcut. Shortcuts are fast and untrustworthy under adversary literacy; trace is slow and grep-able.

Figure 20.2 — DNS lies poison packet sentences — trace from root.

Claim: NEXUS packet field — local jsonl perimeter

`field-dns.py` implements loopback-first binds per seed. `dns-threat-guard` rate limits and permanent blocks abuse. `dns-egress-integrity` hashes answers so tamper in flight is visible. `dns-service-takeover` waits for healthy loopback DNS before steering `resolv.conf`.

`nft` blocks foreign resolver egress when phase = `primary`. Queen FieldFox profiles point DNS at `127.0.0.1` only — navigation without Truth DNS is off-covenant.

Phase 2 roadmap: native resolver (hickory/unbound), DNSSEC validate, TCP + EDNS. Today stub counters document DNSSEC posture honestly — not wire validation theater.

DNS admin portal — no hardcoded production keys

Set `NEXUS_DNS_ADMIN_PASSKEY` in production. Use `NEXUS_DNS_ADMIN_REQUIRE_ENV=1` to refuse seed passkeys entirely.

Ports 7, 77, 777 remain read-only information surfaces — no remote controls without operator covenant. Cleartext admin on WAN is retired vulnerability; localhost bind default.

Admin portal is not the panel at 9477 — it is DNS-specific tooling. Do not conflate them in runbooks.

Field DHCP v3 — issue only, verify always

`field-dhcp/v3` in `field-dhcp.py` binds LAN IP detected via `ip -4 -o addr` unless explicit `NEXUS_FIELD_DHCP_BIND`. Public mode `NEXUS_FIELD_SERVICES_PUBLIC=1` may bind 0.0.0.0 — explicit only.

Figure 20.3 — Issue leases only after sovereign verify — fail closed.

Option 50 requested IP must match lease for MAC — REQUEST without pool match rejected in v2026. Ping conflict check before OFFER — no duplicate static surprises.

DNS option 6 points to Truth Resolver: 127.0.0.1 default, IPv6 ::1. Optional MAC allowlist from seed `field-services-2026-seed.json`.

Lease file: `field-dhcp-leases.json`. Events: `field-dhcp-events.jsonl`. Panel cache: `field-dhcp-panel.json`. Discover rate limits curb starvation attacks.

DHCP is not DNS. Issuing a lease without Truth DNS option pushes clients to foreign resolvers — veto that configuration in peer review.

Sovereign NTP — `field-ntp-2026.py`

Two time layers cooperate: `sovereign-time.py` UDP 9123 signed pulses with micron witness; `field-ntp-2026.py` UDP 123 NTP mode-4 replies gated on sovereign pulse health.

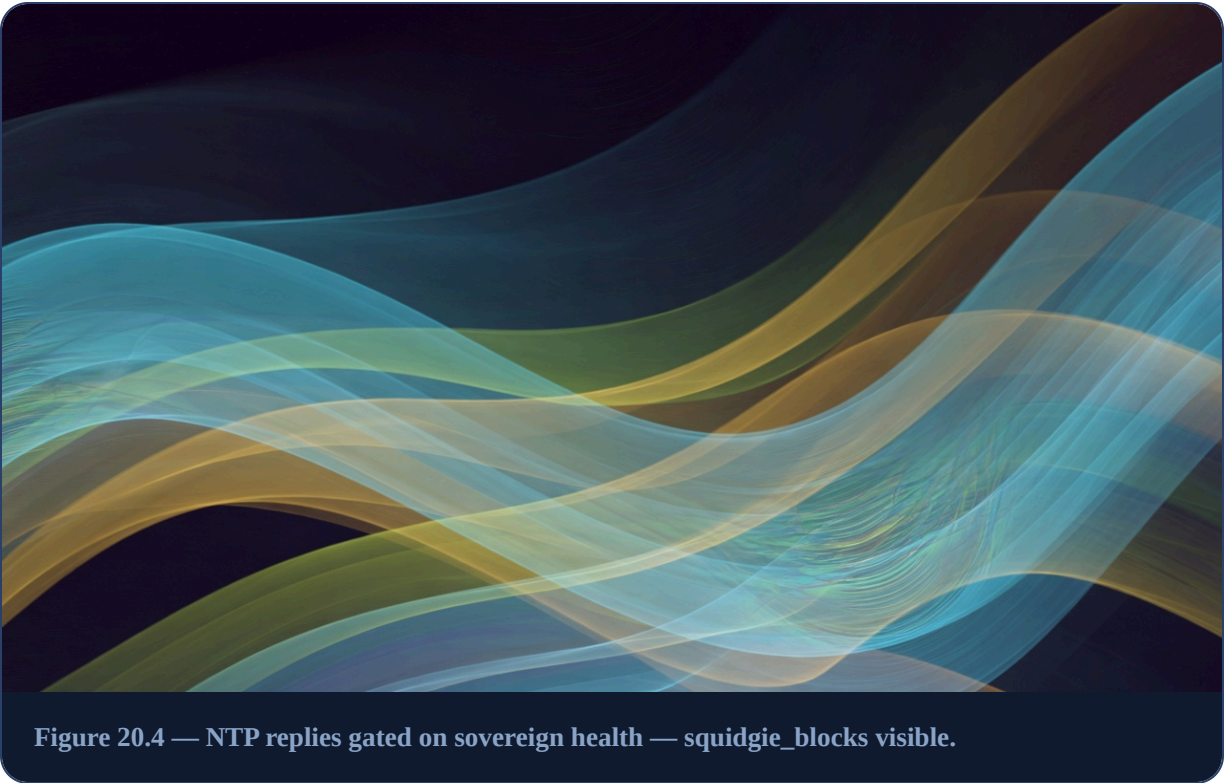


Figure 20.4 — NTP replies gated on sovereign health — `squidgie_blocks` visible.

Daemon starts both when `NEXUS_SOVEREIGN_TIME=1` and `NEXUS_FIELD_NTP=1`. `squidgie_blocks` counter increments when sovereign cache dirty — clients should see failure, not silent pool drift.

Stratum defaults: 2 on normal nodes; 1 when `NEXUS_LAST_HOST=1`. Rate limits on NTP requests mirror DNS discipline.

`grok_world.sh` flow 20 sovereign-first disables pool NTP when `NEXUS_SOVEREIGN_TIME_FIRST=1`. Chrony hook optional via `NEXUS_SOVEREIGN_CHRONY_CONF=1`.

field-services-2026.py — unified panel manifest

Module at `lib/field-services-2026.py` builds `field-services-2026-panel.json` under state dir. Imports slices from `field-dns`, `field-dhcp`, `field-ntp-2026`, `sovereign-time`, `ellie-last-host`.

Posture block reports `loopback_dns_default`, `dhcp_lan_only`, `sovereign_first`, `pool_ntp_fallback`, `last_host` flags from environment.

`vulnerabilities_retired` array is first-class panel data — show it to auditors. edition motto: `loopback-first`, operator-owned, verify at receive.

Commands: `build` refreshes panel; `json` prints cached or rebuilt doc.

ELLIE Last Host — sole survivor mode

From `ELLIE.hpp` Captain Ellie / `TotalTime`: when `NEXUS_LAST_HOST=1`, operator node becomes global DNS, DHCP, TIME on all interfaces. Takeover jumps to `primary` — no waiting while world is gone.

`ellie-last-host.py seal` at daemon boot seals genesis. `posture` reports binds, pool, gateway. `verify` entropy check pairs with sovereign SQUIDGIE `grep`.

Service registry: DNS :53 · DHCP :67 · NTP :123 · Sovereign :9123 · Panel :9477 · Gateway (DHCP opt 3) · `grep` discipline.

Motto in seed: Only computer left — we are DNS, DHCP, TIME, and the other important shit. Treat as operational scripture, not marketing.

`NEXUS_LAST_HOST=1 ./nexus.sh restart` is covenant act. Document who may invoke it and when.

Receive-verify chain across services

DNS: hash answers at egress integrity layer. DHCP: verify MAC, option 50, ping before OFFER. Time: HMAC pulse, monotonic forward, micron witness, SQUIDGIE verdict.

The chain is deliberate parallelism — adversaries should not pick weakest service to squidgie the perimeter. One verify ethic, three ports.

Panel threat tab merges `services_2026` slice with `field-dns/v3` so operator sees retired vulns and live posture in one `grep` session.

Queen and FieldFox inheritance

Queen browser requires Truth DNS lock — no Google shortcut. Field DHCP must hand clients 127.0.0.1 DNS option or Queen navigation lies.

Sovereign time witness timestamps packet field sentences per navigation. Public services chapter is prerequisite for Queen chapter — do not skip.

Opt-in WAN exposure

NEXUS_FIELD_SERVICES_PUBLIC=1 opts into WAN binds documented in seed public_exposure block. dns_wan_default, dhcp_wan_default, ntp_wan_default remain false in seed.

Explicit env is honesty label. Shipping open resolvers by default was retired vulnerability class.

Phase 2 — honest roadmap

DNSSEC wire validation: planned — stub counters today. DHCPv6: schema reserved. Native resolver swap: documented. TCP/EDNS path: phase 2.

Do not teach phase 2 as shipped. Teach it as labeled roadmap beside implemented loopback-first core.

Week-three operator lab — full stack bring-up

Start NEXUS with field services env vars. Run `python3 field-services-2026.py json` — confirm loopback DNS, LAN DHCP, sovereign_first true.

Dig through Truth Resolver locally. DHCP discover test client or vm net. NTP query against 127.0.0.1; confirm stratum traces sovereign.

Open panel :9477 DNS tab — vulnerabilities_retired visible. Archive json slice weekly.

Troubleshooting

resolv.conf steered before loopback DNS healthy: dns-service-takeover waits — check field-dns health first.

DHCP OFFER missing: ping conflict or option-50 mismatch — read field-dhcp-events.jsonl.

NTP silent: check sovereign cache and squidgie_blocks — fix time before blaming clients.

Honest rocks

Loopback DNS + takeover: **Implemented** . DHCP v3 security checks: **Implemented** . Sovereign NTP on 123: **Implemented** .

DNSSEC wire validation: **Phase 2** stub counters. DHCPv6: **Schema reserved** .

Truth DNS — dig +trace walkthrough

Operator runs `dig +trace example.com @127.0.0.1`. Answers should chain from root hints through TLD to authoritative — no shortcut to public resolver IP.

Trace slowness is feature — each hop is logged. Speed from Google DNS is retired vulnerability when trace is required for truth.

dns-threat-guard — abuse classes retired

ANY query floods, per-query fork bombs, unbounded recursion depth — rate limits and takeover gating. Adversaries used DNS as amplifier; Truth Resolver refuses ANY by policy where configured.

Permanent blocks accumulate in threat state — `grep permanent` alongside SQUIDGIE in weekly ops review.

dns-egress-integrity — answer hashing

Hashes answers at egress so in-flight tamper between resolver and consumer is visible. Pair with packet field when suspicious CDN answers appear only on one host.

dns-service-takeover — resolv.conf steering

Takeover waits until loopback DNS healthy — prevents steering clients to broken resolver during boot race. Boot order: `field-dns` healthy, then takeover, then `dhcp` hands leases pointing at 127.0.0.1.

nft foreign resolver block — primary phase

When phase equals primary, nft blocks egress to foreign resolver IPs — forces Truth path.
Disabling nft without disabling foreign resolvers in browser profiles reopens shortcut class.

Field DHCP — pool design and gateway option

Default pool 192.168.50.100–200 per seed. Gateway option 3 set from detected LAN topology on last-host. Document static reservations in seed mac_allowlist when enabled.

Lease seconds default 3600 — shorten on hostile LAN segments if you must.

Field DHCP — DISCOVER rate limits

Discover rate max 12 per window curbs starvation — rogue client cannot exhaust pool before operator notices events jsonl.

Field DHCP — MAC allowlist optional

mac_allowlist_enabled false by default — enabling requires seed maintenance. Use on equipment room segments where only known devices may lease.

field-dns.py and panel integration

field-dns slice in field-services-2026 panel shows binds, phase, health. Merge with threat panel DNS tab for vulnerabilities_retired visibility.

field-ntp-2026 — mode-4 reply anatomy

RFC 5905 mode-4 responses built from sovereign-sourced time. Stratum field reflects operator hierarchy — not pool stratum fiction.

Rate max 30 per window default — prevents NTP amplification abuse on exposed interfaces if public mode ever enabled.

Sovereign-first versus pool fallback

NEXUS_POOL_NTP_FALLBACK=0 default. Opt-in fallback is honesty label — document why fallback exists if enabled.

Chrony hook optional

NEXUS_SOVEREIGN_CHRONY_CONF=1 generates conf pointing at signed source — for sites already on chrony discipline.

ELLIE Last Host — service registry deep dive

DNS 53, DHCP 67, NTP 123, Sovereign 9123, Panel 9477, gateway via DHCP, grep discipline — sole survivor runs full public service stack globally when LAST_HOST set.

ellie-last-host.py seal at boot ties TotalTime genesis metaphor to perimeter daemon start — covenant event logged to ellie-last-host-thermo.jsonl.

ELLIE concepts mapped to services

TotalTime genesis maps to seal command. Session entropy verify maps to SQUIDGIE and apocalypse markers. LOG_THERMO maps to thermo jsonl grep. FieldSocket sealed_time maps to sovereign pulse feeding NTP stratum.

Admin portal ports 7, 77, 777

Read-only information surfaces — not remote control plane. Localhost bind default. TLS panel at 9477 preferred for interactive ops.

NEXUS_DNS_ADMIN_PASSKEY production

Never ship production with seed passkey. NEXUS_DNS_ADMIN_REQUIRE_ENV=1 refuses seed — forces operator to set env in deployment.

Queen FieldFox DNS lock inheritance

Queen requires Truth DNS — browser profile locks 127.0.0.1. Public services chapter is prerequisite: if DHCP hands 8.8.8.8, Queen lock is defeated at lease.

Case study — rogue DHCP on conference Wi-Fi

Without ping conflict and option-50 checks, rogue server could hand bad DNS. Field DHCP v3 retires that class on your segment when you are primary. Document when you are not primary — grep remains essential.

Case study — pool NTP step after resume

Laptop resumes, pool steps clock, sovereign sync returns SQUIDGIE. Fix pool disable, re-sync sovereign, then allow NTP clients. Teaches ordering: sovereign before serving others.

field-services-2026-seed.json as contract

Seed is contract with auditors — vulnerabilities_retired, public_exposure false defaults, receive_verify list mirrors Chapter 19 verify ethic.

DHCPv6 schema reserved

IPv6 DHCP path documented as reserved — do not teach as shipped. DNS option v6 ::1 in seed for forward compatibility.

TCP EDNS DNS phase 2

UDP-only DNS retired as sole story eventually — phase 2 adds TCP and EDNS without changing loopback-first ethic.

Unified threat panel DNS tab

Merges field-dns/v3 with services_2026 slice — one view for DNS DHCP NTP sovereign status and posture flags.

Receive-verify list from seed

HMAC pulse signature, monotonic never backward, realtime skew bound, micron witness, SQUIDGIE grep — same list as Chapter 19, repeated here because services implement it together.

Equipment room segment

MAC allowlist and static reservations for lab gear — Field DHCP events jsonl becomes BOM audit trail.

grep discipline across three ports

Weekly: grep SQUIDGIE, grep dhcp REJECT, grep dns permanent block. Archive field-services-2026-panel.json with receipts.

Long-form primer — public services as one perimeter

Chapter 20 is not a DNS chapter, a DHCP chapter, and a time chapter bound in one file for convenience. It is one perimeter ethic expressed on three ports that every downstream consumer — Queen browser, FieldFox, ellie-last-host survivor, generic LAN clients — must inherit coherently. If DNS is truth on loopback but DHCP hands Google DNS, you built a lie into lease option 6. If DHCP is pristine but NTP answers from pool while sovereign is SQUIDGIE, you trained clients to trust a stepped clock. Coherence is the product. Individual daemons are components.

Truth DNS begins with refusal: refusal to take resolver shortcuts, refusal to bind WAN by default, refusal to steer resolv.conf before loopback resolver is healthy, refusal to treat ANY queries as unlimited entertainment for amplifiers. dig +trace from root is slow because it is doing real work — walking delegation chain you can log. When operators complain about latency, answer with terror-threat literacy: shortcuts are where adversaries hide answers you never traced. Rate limits and dns-threat-guard permanent blocks are not annoyance; they are retired vulnerability classes given names in field-services-2026-seed.json.

dns-egress-integrity hashing closes the gap between resolver output and application belief. Two hosts querying same name should be able to compare hashes when suspicion arises. Pair hashes with packet field captures when CDN answers diverge — you are practicing verify-at-receive for DNS same as for time.

Field DHCP v3 is issue-only, verify-always made concrete: bind LAN IP detected or explicit, not 0.0.0.0 unless NEXUS_FIELD_SERVICES_PUBLIC=1 documents WAN exposure. Option 50 requested address must match pool assignment for MAC — stops REQUEST spoofing that

grabbed arbitrary addresses. Ping conflict before OFFER — stops silent duplicate IPv4 on segment. DNS option 6 toward 127.0.0.1 — steers clients back to Truth Resolver you actually run. Events jsonl is forensic backbone for lease denials — grep REJECT and CONFLICT weekly.

MAC allowlist optional remains off by default because many home labs rotate devices — enabling allowlist shifts operational burden to seed maintenance. Equipment room segments with fixed BOM are the sweet spot for allowlist true. Document static reservations in seed when printers or scopes must never float.

Sovereign NTP layer two — field-ntp-2026 — exists because the world still speaks NTP on 123. Sovereign-first does not mean every client suddenly speaks UDP 9123 JSON pulses. It means mode-4 replies are gated on sovereign health — squidgie_blocks counter visible in panel. Clients that ignore blocking and drift are misconfigured; fix clients, do not disable gate to comfort them.

field-services-2026.py unify panel slice is operator single pane for auditors. vulnerabilities_retired array is not shame list — it is accomplishment list of classes Grok rewrite 2026 refuses to ship again. Show it to anyone asking whether your perimeter is default insecure. posture block booleans loopback_dns_default, dhcp_lan_only, sovereign_first, last_host — grep env and compare to panel; mismatch means stale panel or wrong daemon env.

ELLIE Last Host is extreme covenant posture — NEXUS_LAST_HOST=1 — sole survivor serves global DNS DHCP TIME on all interfaces, takeover jumps primary without waiting for world that is gone. ellie-last-host.py seal at boot, posture command reports binds pool gateway, verify checks entropy covenant. This is not daily driver config. It is documented exception with motto in seed: only computer left. Operators who enable it without tabletop exercise are cosplaying apocalypse. Run drill 20-E.

Admin portal security — NEXUS_DNS_ADMIN_PASSKEY, NEXUS_DNS_ADMIN_REQUIRE_ENV=1 — closes hardcoded seed passkey production leak. Ports 7 77 777 read-only information, localhost bind, TLS panel 9477 preferred. Teach difference between DNS admin portal and threat panel — conflating them causes wrong escalation paths.

Queen inheritance closes loop to Chapter 21: Truth DNS lock in FieldFox profile requires DHCP option 6 truth. Sovereign time witness timestamps navigation. Public services are prerequisite, not optional elective, for Queen READY verdict in honest deployments.

Phase 2 honesty — DNSSEC wire validation, DHCPv6, TCP EDNS native resolver — stays labeled roadmap. Stub counters today teach honesty. Do not tell students DNSSEC is fully validated when seed says stub_trace. Do tell them where roadmap lives.

Week-three lab expanded: bring NEXUS with field services env; field-services-2026.py json; dig trace @127.0.0.1; DHCP test vm; ntpdate or chrony query 127.0.0.1; panel DNS tab screenshot with vulnerabilities_retired; archive panel json beside sovereign receipts.jsonl — one zip per week, covenant discipline.

Case study expanded — conference Wi-Fi rogue DHCP: your laptop not primary; Field DHCP protections do not apply on foreign segment; Truth DNS on loopback still protects local resolver path on machine — understand what travels with you versus what protects LAN you administer.

Case study expanded — dual clock laptop: pool chrony and sovereign serve both running; SQUIDGIE intermittent; fix by sovereign-first disable pool; document in runbook; grep chrony.conf and ntp sources.

Receive-verify list from seed echoes Chapter 19 — perimeter implements together: HMAC, monotonic forward, skew bound, micron witness, SQUIDGIE grep. Auditors love repeated ethic across chapters — it is not redundancy, it is enforceable policy.

Opt-in WAN NEXUS_FIELD_SERVICES_PUBLIC=1 — explicit env, seed public_exposure all false defaults — never ship open resolver DHCP NTP to internet by default again. If you must expose, pair with rate limits, monitoring, and written risk acceptance.

Service-by-service operator runbook

Truth DNS daily: confirm bind 127.0.0.1; dig +trace test; check dns-threat-guard blocks; verify resolv.conf points loopback after takeover; grep permanent blocks. Field DHCP daily: confirm LAN bind not accidental 0.0.0.0; spot-check events jsonl; verify option 6 is 127.0.0.1; ping conflict test monthly. Sovereign NTP daily: sovereign-time sync USER_OK; field-ntp-2026 squidgie_blocks zero; stratum matches policy; pool fallback env documented if on.

Field DNS implementation notes for auditors

field-dns.py exposes panel slice consumed by field-services-2026.py. Binds ipv4 default 127.0.0.1 and 127.0.0.53 per seed — dual loopback story for systemd-resolved coexistence on some distros. enforce_resolv true means takeover discipline is not optional theater. block_foreign_resolvers true pairs with nft when primary. reject_any_queries true retires ANY amplification class. max_udp_payload 1232 follows EDNS sanity. dnssec_validation stub_trace honestly admits wire validation is phase 2 — auditors should see stub counters, not marketing checkmarks.

Field DHCP implementation notes for auditors

field-dhcp.py schema v3 writes leases json, events jsonl, panel cache. Discover rate limit 12 per 60s window. Lease seconds from env default 3600. Pool from env or seed.

require_takeover_primary means rogue WAN DHCP default blocked — LAN administrator must assert primary phase. ping_conflict_check true before OFFER. option 50 validation rejects REQUEST that does not match assigned pool IP for MAC. DNS servers v4 and v6 from env with Truth defaults. These are grep-able behaviors — read events jsonl after test VM discover.

field-ntp-2026 implementation notes for auditors

Module gates mode-4 replies on sovereign module health via import of sovereign-time.py. squidgie_blocks stat increments on dirty cache — panel must show it. BIND default 127.0.0.1 — WAN requires public mode. STRATUM from env with last-host elevation. Rate limit 30 per 60s on requests. NTP_EPOCH conversion uses standard 1900 epoch offset. PID and lock files under state dir — standard daemon hygiene.

ELLIE Last Host tabletop script

Read seed ellie_last_host block aloud in team meeting. List sole_global_services: dns, dhcp, time_ntp, time_sovereign, gateway, panel, dns_admin, grep. Assign human owner for NEXUS_LAST_HOST=1 invocation. Dry-run ellie-last-host.py posture without restart. Document gateway DHCP option 3 behavior. Pre-write customer communication if last host is production continuity plan — not surprise toggle.

Cross-chapter integration table

Consumer	Needs from Ch 20	Failure symptom
Queen FieldFox	Truth DNS 127.0.0.1, DHCP option 6	Foreign resolver shortcut, broken honorability DNS
AMOURANTHRTX logs	Sovereign USER_OK, NTP gated	Mis-ordered thermo vs network events
GPS precision	Stable UTC from sovereign chain	Pretty dots, bad correlation
Threat panel	field-services-2026.json fresh	Stale posture flags
Last host survivor	ALL services primary	World gone but still waiting takeover

Retired vulnerabilities — expanded commentary

Hardcoded DNS admin passkeys in seed only — production must use env passkey or Hostess7 require env flag. DHCP bind 0.0.0.0 without takeover primary — blocked by default so rogue server cannot own segment silently. Pool NTP as sole authority — sovereign-first policy replaces blind pool trust. dig-per-query fork bomb — rate limits and takeover gating retained until native resolver swap. DHCP REQUEST without option-50 pool match — rejected in v2026 to stop arbitrary address grabs. No ARP ping conflict detect — ping probe before OFFER now mandatory story. UDP DNS only — TCP EDNS DNSSEC documented phase 2. Cleartext admin on 7 77 777 — localhost bind default, TLS panel preferred. Each retired item should appear in operator training slides — not buried only in seed json.

Service	Port	2026 default	Retired vulns
Truth DNS	53 UDP/TCP	127.0.0.1 / ::1 only	Foreign resolver shortcut, ANY flood, dig fork bomb
Field DHCP	67 UDP	LAN IP bind, takeover primary only	0.0.0.0 rogue DHCP, option-50 mismatch
Sovereign NTP	123 UDP	Operator stratum from signed pulses	Pool NTP as sole authority

```
python3 field-services-2026.py json
python3 sovereign-time.py sync
python3 ellie-last-host.py posture
python3 ellie-last-host.py verify
NEXUS_LAST_HOST=1 ./nexus.sh restart
```

Operator drills

Drill 20-A — Truth trace. dig +trace example domain through 127.0.0.1 only. Goal: no shortcut resolver in path.

Drill 20-B — DHCP conflict. Place static IP in pool range; attempt lease. Goal: ping conflict blocks OFFER; event in jsonl.

Drill 20-C — Sovereign NTP gate. Query NTP; induce SQUIDGIE; confirm squidgie_blocks increments.

Drill 20-D — Panel slice. Archive field-services-2026-panel.json; verify vulnerabilities_retired matches seed.

Drill 20-E — Last host tabletop. Document who sets NEXUS_LAST_HOST=1; run posture dry-run without WAN.

Study questions

1. Why is NEXUS_FIELD_SERVICES_PUBLIC=1 explicit opt-in?
2. What does dns-service-takeover wait for before steering resolv.conf?
3. Explain option-50 validation and why it matters.
4. How do sovereign-time.py and field-ntp-2026.py cooperate?
5. What services does ELLIE Last Host registry enumerate?
6. Name three retired vulnerabilities from 2026 seed.
7. Why must DHCP option 6 point at Truth Resolver for Queen?
8. What is honest label for DNSSEC today?
9. How does nft foreign resolver block relate to primary phase?
10. What unified command refreshes the services panel slice?

Chapter 21 — Field Browser Queen →

Truth DNS and Field DHCP must be restarted together after seed policy changes — panel slice stale until field-services-2026.py build runs.

dig +trace failures on offline laptop are expected — Truth Resolver cannot walk to root without uplink. Distinguish offline from misconfiguration by checking bind and takeover phase first.

DHCP option 6 multiple DNS servers should still list 127.0.0.1 first — secondary forwarders are policy choice, not Google defaults.

field-dhcp-events.jsonl rotation at EVENTS_LOG_MAX prevents unbounded disk — archive rotated logs before they are truncated.

NTP rate limits protect against accidental amplification if operator enables public mode — keep rate env defaults unless measurement proves need.

ellie-last-host-thermo.jsonl pairs covenant events with thermo ethic — grep alongside sovereign receipts during last-host drills.

DNS admin portal passkey rotation should follow same ceremony as sovereign key rotation — documented operator root events.

Threat panel merge of field-dns/v3 and services_2026 means one tab for auditor tour — practice demo path before external review.

MAC allowlist enablement without updating seed is silent failure — always version control seed json with DHCP policy.

Ping conflict detect may fail if ICMP blocked on segment — document segment exceptions where ARP-only checks are insufficient.

Gateway option 3 on last host must match actual router story — wrong gateway is DHCP lie as much as wrong DNS.

IPv6 ::1 DNS option in seed prepares dual-stack Truth path — do not disable IPv6 loopback bind without reading field-dns panel.

Stub DNSSEC counters exist to honesty-label phase 2 — teach auditors difference between trace integrity and chain validation.

field-services-2026 edition string Grok rewrite 2026 is version anchor — cite in compliance packets.

Receive-verify list in seed is copy-paste of Chapter 19 policy — intentional duplication for auditors who read only services seed.

Equipment room static reservations reduce DHCP entropy on factory floors — events jsonl becomes asset audit.

Cleartext admin ports 7 77 777 on localhost still require passkey hygiene — localhost is not automatic trust of all local users.

NEXUS_FIELD_SERVICES_PUBLIC=1 requires written WAN exposure approval — attach to operator covenant filings.

Queen browser FieldFox profile DNS lock must be re-exported after FieldFox upgrade — upgrades sometimes reset prefs.

Weekly archive zip: field-services panel json, sovereign receipts, dhcp events tail, dns threat blocks — minimum compliance habit.

Truth DNS and Field DHCP must be restarted together after seed policy changes — panel slice stale until field-services-2026.py build runs.

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ellie-last-host-thermo.jsonl pairs covenant events with thermo ethic — grep alongside sovereign receipts during last-host drills.

DNS admin portal passkey rotation should follow same ceremony as sovereign key rotation — documented operator root events.

Threat panel merge of field-dns/v3 and services_2026 means one tab for auditor tour — practice demo path before external review.

MAC allowlist enablement without updating seed is silent failure — always version control seed json with DHCP policy.

Ping conflict detect may fail if ICMP blocked on segment — document segment exceptions where ARP-only checks are insufficient.

Gateway option 3 on last host must match actual router story — wrong gateway is DHCP lie as much as wrong DNS.

IPv6 ::1 DNS option in seed prepares dual-stack Truth path — do not disable IPv6 loopback bind without reading field-dns panel.

Stub DNSSEC counters exist to honesty-label phase 2 — teach auditors difference between trace integrity and chain validation.

field-services-2026 edition string Grok rewrite 2026 is version anchor — cite in compliance packets.

Receive-verify list in seed is copy-paste of Chapter 19 policy — intentional duplication for auditors who read only services seed.

Equipment room static reservations reduce DHCP entropy on factory floors — events.jsonl becomes asset audit.

Cleartext admin ports 7 77 777 on localhost still require passkey hygiene — localhost is not automatic trust of all local users.

NEXUS_FIELD_SERVICES_PUBLIC=1 requires written WAN exposure approval — attach to operator covenant filings.

Queen browser FieldFox profile DNS lock must be re-exported after FieldFox upgrade — upgrades sometimes reset prefs.

Weekly archive zip: field-services panel.json, sovereign receipts, dhcp events tail, dns threat blocks — minimum compliance habit.

Truth DNS and Field DHCP must be restarted together after seed policy changes — panel slice stale until field-services-2026.py build runs.

dig +trace failures on offline laptop are expected — Truth Resolver cannot walk to root without uplink. Distinguish offline from misconfiguration by checking bind and takeover phase first.

DHCP option 6 multiple DNS servers should still list 127.0.0.1 first — secondary forwarders are policy choice, not Google defaults.

field-dhcp-events.jsonl rotation at EVENTS_LOG_MAX prevents unbounded disk — archive rotated logs before they are truncated.

NTP rate limits protect against accidental amplification if operator enables public mode — keep rate env defaults unless measurement proves need.

ellie-last-host-thermo.jsonl pairs covenant events with thermo ethic — grep alongside sovereign receipts during last-host drills.

DNS admin portal passkey rotation should follow same ceremony as sovereign key rotation — documented operator root events.

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Gateway option 3 on last host must match actual router story — wrong gateway is DHCP lie as much as wrong DNS.

IPv6 ::1 DNS option in seed prepares dual-stack Truth path — do not disable IPv6 loopback bind without reading field-dns panel.

Evidence anchor — grep and sources

Major claims in this chapter anchored for reproducibility. **Implemented** = grep today;

Metaphor = intuition; **Philosophy** = discipline.

Claim	Statement	Label	Evidence
Truth DNS	127.0.0.1 loopback	Implemented	field-dns.py
Field DHCP	Issue + verify always	Implemented	field-dhcp.py v3 schema
dig +trace	Root trace discipline	Implemented	No resolver shortcut
DNSSEC marketing	Full wire validation	Metaphor	stub_trace honest phase 2

OFFER after ping_conflict_check; option 6 → loopback DNS

```
python3 field-services-2026.py json
```

Source paths

- NEXUS-Shield/field-dns.py
- NEXUS-Shield/field-dhcp.py
- NEXUS-Shield/field-services-2026.py

Chapter summary — before you turn the page

Loopback-first public services — Truth DNS, Field DHCP, Sovereign NTP — WAN only with explicit operator flag.

Field Browser Queen — Hold All Gates

Introduction — Queen doctrine

Nothing optional. Hold all gates. MP4. We want it ALL. Chapter 21 is Field Browser Queen — the complete web surface carried into the packet field without amputating capabilities to feel safe.

Figure 21.1 — Hold all gates: WebRTC through gatekeeper, MP4 in-tree.

Claim: QUEEN_READY — all gates held in-engine

On the way — what you will learn

Queen holds all gates — WebRTC through gatekeeper, MP4 mandatory in-tree, EME held not omitted. On the way you will link browser perimeter to AMOURANTHRTX spine, KILROY sovereign field, and packet-field receipts.

On the way — Queen browser

- QUEEN_BROWSER_BUILD and rtx() spine
- Nothing optional security posture
- Thermo per WebGL context
- Sovereign packaging with KILROY

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| 2. Gate mode — hold, never omit | in-tree |
| 3. WebGL, WebGPU, and thermo per context | 6. EME / Widevine — held, not omitted |
| 4. WebRTC — gated realtime | 7. Service Workers, WASM, storage gates |
| | 8. Sensors, devices, UI gates |

- | | |
|---|---|
| 9. FieldFox — ship now engine | 12. KILROY and field package — sovereign field |
| 10. queen-browser — in-engine RTX path | 13. Millennium track — Ladybird and Servo |
| 11. NEXUS stack binding | 14. Panel slice and queen_verdict |

Reality is 3D · Time is linear · Energy can be moved. Queen does not shrink the web. Queen holds gates: every capability exists; every wire exit earns a receipt.

Wrong posture: disable WebRTC for privacy. Right posture: WebRTC flows through Connection Gatekeeper with honorability and packet field. Wrong: MP4 optional pack. Right: MP4 + H.264 + AAC mandatory in-tree.

Gate mode — hold, never omit

Capabilities are not toggles that remove surface area. They are gates the operator holds via Connection Gatekeeper, Packet Field, Honorability DB, Fair Ad Guardian, Thermal Governor, Truth DNS, and Sovereign Time witness.

`field-queen-gates-seed.json` lists every gate with `held:true` — `html_dom`, `javascript`, `wasm`, `webgl`, `webgpu`, `mse`, `eme`, `webrtc`, `websockets`, `camera`, `microphone`, `legacy_plugins_surrogate`, and more.

`field-queen-browser.py` computes `queen_verdict`: `QUEEN_READY` only when all gates held, `mp4_mandatory true`, `hold_all_gates` env on.

WebGL, WebGPU, and thermo per context

WebGL 1/2 and WebGPU stay on. Each GPU context bills thermo via Thermal Governor — energy honesty per tab, not global hand-wave.

Disabling GPU APIs to feel safe amputates the web Queen promises to carry. Hold gates instead: score egress, receipt entropy, throttle abusive contexts.

Canvas 2D shares gpu layer gates — not a second-class citizen.

WebRTC — gated realtime

WebRTC audio, video, and data channels stay on. Gatekeeper scores each peer connection. Camera and microphone gates require operator consent layer — held, not removed.

Figure 21.3 — WebRTC egress through Connection Gatekeeper — no silent holes.

STUN/TURN flows appear in packet field sentences. Honorability precedes sustained egress. A peer without trust story does not get silent long-term hole punching.

MSE, MP4, H.264, AAC — mandatory in-tree

Media Source Extensions required. Containers include video/mp4, video/webm, audio/mp4, audio/mpeg. Video codecs: avc1, h264, hev1, vp9, av01. Audio: mp4a, aac, opus, flac.

Path: MSE → Gecko media → FFmpeg in-tree. MP4 is not optional pack. queen_verdict returns MP4_MISSING if video/mp4 absent from container list.

Tell users to install Chrome for video is retired vulnerability class. FieldFox ships full codec tree.

EME / Widevine — held, not omitted

Encrypted Media Extensions stay on. DRM is held — gatekeeper scores license server flows. Omitting DRM does not remove content sites; it lies about capability surface.

Fair Ad Guardian and honorability apply to license traffic same as ads and analytics.

Service Workers, WASM, storage gates

Service Workers and Cache API on — entropy receipts for persistent script surfaces. WASM on — same gate discipline as JS.

IndexedDB, Web Storage, File System Access, SharedArrayBuffer with COOP/COEP — all held. Storage without receipts is exfiltration runway.

Sensors, devices, UI gates

Geolocation, Notifications, Push, Web Bluetooth, WebUSB, Web Serial, Gamepad — held with consent and gatekeeper scoring.

Popups, window.open, third-party iframes — held. Legacy Flash/Java/Silverlight via WASM surrogates — surface preserved, plugin VM gone.

PDF inline, SVG, MathML — document layer held for scholarly web.

FieldFox — ship now engine

FieldFox is hardened Gecko fork: telemetry stripped, Truth Resolver DNS lock at 127.0.0.1, NEXUS native messaging bridge, full codec tree.



Figure 21.4 — FieldFox: hardened Gecko, Truth DNS, full codec tree.

`NEXUS_FIELD_BROWSER_QUEEN=1 ./lib/fieldfox-launch.sh` starts Queen profile.
`python3 lib/field-queen-browser.py json` returns `queen_verdict QUEEN_READY`
when posture correct.

Process names tracked include `fieldfox`, `firefox`, `chromium` family for browser-awareness integration.

queen-browser — in-engine RTX path

`queen-browser` binary — AMOURANTH RTX Vulkan + SDL3 inside. In-engine UI via `FieldWebPanel`. `QueenBoot.comp` — 2026 aqua/rose chrome. Hostess 7 Forever Watchguard in 50/50 split deck — the in-process robot-brain stack inside Queen (**Metaphor** for gate+recall architecture; not a government deliverable), not a cloud tab.

Figure 21.2 — One rtx() spine: engine and browser faces.

Brain imaging: Hostess folds 1000–1200 word textbook beats into SDF plates (signed distance fields are free — procedural topology, not megapixel rent). Series-of-series neural nets recall plates via vision OCR; grep blocks land in SDL text documents. Five decisions per segment: integrate · reimage · toss · sdl_store · sdf_plate. Teach: `./Hostess7.sh sdf-teach seed` · segment: `./Hostess7.sh sdf-segment <file>`.

`NEXUS_INSTALL_ROOT=Queen ./build/rtx/bin/Linux/queen-browser --sovereign --queen` links engine spine when `QUEEN_BROWSER_BUILD` set.

Two ship tracks coexist: FieldFox Gecko now; queen-browser Vulkan in-engine for sovereign RTX field deployments.

NEXUS stack binding

Navigation → Packet Field → Gatekeeper → Honorability → Thermo receipt. Upstack: Truth DNS (no Google shortcut). Upstack: Sovereign Time (SQUIDGIE witness).

`fieldfox-native-bridge.py` native messaging to gatekeeper. `browser-awareness.py` live tab honorability feeds panel.

`connection-gatekeeper.py` 10-axis intent scoring per flow. packet field jsonl per navigation.

KILROY and field package — sovereign field

Queen Forge packages kernel + browser + secure stack in `field/sovereign/`. KILROY Field OS (7.1.1-kilroy) provides `/proc/kilroy_field` at syscall boundary — `CONFIG_RTX_FIELD_DIE` field die in kernel.

Userspace Queen runs on top — one field, sealed manifest, Grok Build secure channel inside. Kernel FCC parallel (Chapter 9) shares vocabulary with userspace CFL.

field package is not zip convenience — it is sealed operator manifest: kernel image, Queen browser, Truth DNS, Field DHCP, sovereign time, panel slice, grep discipline documented.

Year 3000 vision: one sovereign stack — Queen browser + NEXUS + Truth DNS + sovereign DHCP/TIME + ELLIE Last Host. Web does not shrink; gates do not open without receipts.

Millennium track — Ladybird and Servo

Ladybird / Servo independent engines inherit field-queen-gates-seed.json — same gate doctrine, no Chromium dependency, no telemetry chew.

FieldFox ships first. Millennium engines are not excuse to delay gates — they are future trunks with same held manifest.

Panel slice and queen_verdict

Threat panel merges field_queen_browser slice: gates.all_held, codecs.mp4_mandatory, queen_verdict, browser_awareness honorability.

Verdicts: QUEEN_READY, QUEEN_OFF, GATES_INCOMPLETE, MP4_MISSING, HOLD_ALL_GATES_OFF — grep-able, not decorative.

field-queen-browser.py gate <id> inspects individual gate held state.

Vulnerabilities retired — browser class

Optional DRM/WebRTC/WebGPU toggles retired — Queen holds gates. Chromium telemetry chew retired — FieldFox strips it. Foreign DNS shortcut retired. Pool NTP sole clock retired. MP4 optional pack retired.

Operator week-four lab

Launch FieldFox with Queen env. Confirm Truth DNS lock. Play MP4 in-tree sample. Open WebRTC test page — confirm gatekeeper scores appear in packet field.

Run field-queen-browser.py json — QUEEN_READY. Cross-check sovereign-time status — SQUIDGIE should block clean verdict if time dirty.

Honest rocks

field-queen-gates-seed.json all held: **Implemented** . field-queen-browser.py panel: **Implemented**

. FieldFox launch script: **Implemented** .

queen-browser Vulkan in-engine: **Implemented** build path; operator may ship FieldFox first.
Ladybird production: **Millennium track** .
KILROY on every host today: **Field package path** — QEMU/bare-metal per Chapter 12 rocks.

Gate-by-gate reference — render and script layers

html_dom, javascript, wasm, service_workers — held at script/render boundary. Disabling JS to feel safe is omit posture — rejected.

Gate-by-gate reference — GPU layer

webgl, webgpu, canvas_2d — thermo per context via Thermal Governor. GPU off is not Queen.

Gate-by-gate reference — media layer

mse, eme — MP4 H264 AAC mandatory; Widevine held. License server flows scored like any egress.

Gate-by-gate reference — realtime layer

webrtc — peers gated; STUN/TURN in packet field. Data channels same scoring.

Gate-by-gate reference — network layer

websockets, fetch_xhr, http2_http3 — QUIC flows are first-class scored connections.

Gate-by-gate reference — storage layer

indexeddb, file_system_access, shared_array_buffer — persistence requires entropy receipt story.

Gate-by-gate reference — sensor and device

geolocation, camera, microphone, bluetooth, usb, serial, gamepad — consent plus gatekeeper.

Gate-by-gate reference — document and UI

pdf_inline, svg_mathml, popups, third_party_embeds — scholarly and legacy web surfaces preserved.

field-queen-browser.py API surface

Commands: build, json, gates, codecs, gate id — operator tooling for manifest audit without opening seed by hand.

fieldfox-native-bridge.py

Native messaging connects browser events to connection-gatekeeper.py — every navigation can emit packet field sentence.

connection-gatekeeper.py — ten-axis intent

Scores process path, port habit, direction, payload hints — verdict before sustained egress. WebRTC peers get same axes.

browser-awareness.py — tab honorability

Active tab trust feeds panel browser_awareness block — operator sees honorability live, not after breach.

Fair Ad Guardian — ad egress

Separates first-party from junk — reduces blind blocker omit posture while holding ad gates.

Thermal Governor — per-tab energy

WebGL and WebGPU contexts bill thermo ethic — abusive miner tabs show energy story in receipts.

Honorability DB — site trust

Persistent store — navigation consults before egress proceeds. Queen doctrine: honorability before egress.

Codec path — MSE to FFmpeg

Container video/mp4 required. Video avc1 h264; audio mp4a aac. Path MSE Gecko media FFmpeg in-tree — document in runbooks for support tickets.

EME Widevine — held scoring

License server URLs appear in packet field — operator can archive DRM egress like any other.

Legacy WASM surrogates

Flash Java Silverlight surfaces replaced by WASM — capability preserved, attack surface reduced versus NPAPI era, but gate still held.

FieldFox launch environment

NEXUS_FIELD_BROWSER_QUEEN=1, HOLD_ALL_GATES=1, MP4=1, truth dns lock, connection gatekeeper, packet field, sovereign time — all default on in posture().

queen-browser build flags

QUEEN_BROWSER_BUILD links AMOURANTHRTX spine — Vulkan SDL3 FieldWebPanel QueenBoot.comp — for sovereign RTX deployments without separate Gecko binary.

FieldWebPanel — in-engine UI

Hosts Field Primer and operator pages inside engine — navigation still passes packet field when networked.

QueenBoot.comp chrome

Aqua rose 2026 shader chrome — rhymes with RTX Zero panel aesthetic — visual covenant of Queen edition.

Hostess 7 Forever Watchguard

50/50 split deck motif — watchguard persona beside content — operator reminder that gates are held not hidden.

KILROY CONFIG_RTX_FIELD_DIE

Kernel field die parallels userspace FieldX86Die — FCC entropy feedback vocabulary shared Chapter 9.

proc kilroy_field

Syscall boundary telemetry — userspace Queen reads kernel field state in sovereign package deployments.

field/sovereign/ manifest

Sealed bundle: KILROY image, Queen browser or FieldFox, NEXUS services, Truth DNS, DHCP, time, panel configs, grep runbook.

Queen Forge packaging

Produces field package for bare-metal or QEMU — Grok Build secure channel inside manifest — not casual zip.

Grok Build secure channel

Build integrity inside sovereign package — operator verifies manifest before deploy.

Year 3000 sovereign stack vision

One stack: Queen plus NEXUS plus Truth DNS plus DHCP TIME plus ELLIE Last Host — web full size, gates never open without receipts.

Ladybird Servo millennium

Inherit field-queen-gates-seed.json — no Chromium — same held manifest when production ready.

Chromium family process awareness

field-queen-browser tracks chromium chrome brave edge names for awareness when operators run non-FieldFox binaries — gate doctrine still applies via NEXUS hooks where installed.

vulnerabilities_retired browser list

Optional toggles, telemetry chew, foreign DNS, pool NTP sole, MP4 optional — all retired classes named in seed.

Case study — WebRTC without gatekeeper

Legacy posture disabled WebRTC — sites broke silently. Queen holds WebRTC, scores peers, logs STUN — operator sees hole punch attempt in jsonl.

Case study — MP4 optional pack failure

Sites served H264 only — optional codec pack meant install Chrome. Queen mandatory in-tree — support ticket closes with FieldFox play confirm.

Case study — sovereign SQUIDGIE during browsing

Time dirty — navigation receipts mis-timestamped — honorability timeline wrong. Fix sovereign sync before blaming site malice.

Panel queen_verdict debugging

GATES_INCOMPLETE — inspect seed held flags. MP4_MISSING — codecs block.
HOLD_ALL_GATES_OFF — env wrong. QUEEN_OFF —
NEXUS_FIELD_BROWSER_QUEEN=0.

Operator covenant clause 6 — hold gates

Chapter 18 clause six — every capability exists, every wire exit earns receipt — Queen is covenant implementation.

HTTP/3 QUIC held

Modern transport not omitted — scored flows in packet field — no transport amputation.

COOP COEP isolation

SharedArrayBuffer gate requires policy headers — held with isolation, not disabled by default.

Notifications and Push

Sensor layer gates — push subscriptions egress scored — no silent background omit.

Third-party embeds

Iframes held — Fair Ad Guardian and gatekeeper score embed egress — amputating iframes breaks scholarly web.

MathML SVG scholarly web

Document gates preserve equations and diagrams — Field Primer itself benefits from same doctrine.

NEXUS bindings block in seed

connection_gatekeeper, packet_field, browser_awareness, honorability_db, fair_ad_guardian, thermal_governor, sovereign_time, field_dns_truth_resolver — all true in nexus_bindings.

Long-form primer — Queen as complete web with held gates

Field Browser Queen rejects the privacy-fork lie that amputation equals safety. Amputating WebRTC does not remove WebRTC from the web — it removes your ability to see WebRTC when sites demand it. Amputating MP4 codecs does not remove video — it ships users to Chrome where telemetry chew awaits. Queen holds gates: capabilities remain; egress earns receipts; operator remains final authority at keyboard.

Doctrine table in seed lists thirty gates held true — `html_dom` through `legacy_plugins_surrogate`. `field-queen-browser.py` gate manifest computes `all_held` only when every seed gate reports held. `queen_verdict` `QUEEN_READY` requires `all_held`, `mp4_mandatory`, `video/mp4` in container list, `hold_all_gates` env on, `queen_enabled` on. Any other verdict is actionable: `GATES_INCOMPLETE` inspect seed; `MP4_MISSING` fix codecs; `HOLD_ALL_GATES_OFF` fix env; `QUEEN_OFF` enable queen.

WebGL WebGPU `canvas_2d` share gpu layer — Thermal Governor bills per context. Miners and abusive shaders become energy story, not silent fan spin. Disabling GPU APIs would break scholarly visualizations, maps, and tools — unacceptable amputation.

WebRTC realtime layer stays on — STUN TURN peers appear in packet field; Connection Gatekeeper scores intent per peer; camera microphone gates require consent plus scoring. Disable WebRTC privacy fork posture is explicitly named wrong in wiki and seed `vulnerabilities_retired`.

MSE EME media layer — MP4 H264 AAC mandatory in-tree via MSE Gecko FFmpeg path. EME Widevine held — license servers scored, not omitted. Tell users use Chrome for video is retired class.

Network layer websockets fetch xhr http2 http3 QUIC — all held, all scored. Storage indexeddb `file_system_access` `shared_array_buffer` — persistence without receipts was exfiltration runaway; Queen requires receipt story.

Sensors geolocation notifications push camera microphone bluetooth usb serial gamepad — consent and gatekeeper. UI popups `third_party_embeds` — iframes held with Fair Ad Guardian and embed scoring — scholarly web needs embeds.

Document layer `pdf_inline` `svg_mathml` — equations and diagrams preserved. Legacy plugins Flash Java Silverlight via WASM surrogates — surface preserved, NPAPI VMs gone.

FieldFox ship-now — hardened Gecko, telemetry stripped, Truth DNS 127.0.0.1 lock, native messaging bridge, full codec tree. `fieldfox-launch.sh` with `NEXUS_FIELD_BROWSER_QUEEN=1`. `field-queen-browser.py` json for panel.

queen-browser in-engine — AMOURANTHRTX Vulkan SDL3 FieldWebPanel QueenBoot.comp Hostess7 Watchguard — QUEEN_BROWSER_BUILD path for sovereign RTX field without separate Gecko binary. Two tracks coexist; operators may ship FieldFox first while engine path matures on hardware targets.

NEXUS stack binding prose: Navigation generates packet field sentence; Connection Gatekeeper scores; Honorability DB consults site trust; Fair Ad Guardian classifies ad egress; Thermal Governor bills energy; Truth DNS ensures names resolve without foreign shortcut; Sovereign Time ensures timestamps defendable — SQUIDGIE blocks clean story. fieldfox-native-bridge.py wires browser events to gatekeeper. browser-awareness.py feeds live tab honorability to panel.

KILROY field package — Queen Forge bundles kernel browser services under field/sovereign/. KILROY 7.1.1-kilroy CONFIG_RTX_FIELD_DIE provides proc kilroy_field syscall boundary telemetry. Userspace Queen runs atop — one field sealed manifest Grok Build secure channel inside. Kernel FCC parallel Chapter 9 shares vocabulary with userspace CFL — not duplicate simulation, shared discipline.

Year 3000 vision in FIELD-BROWSER-2026.md — one sovereign stack Queen NEXUS Truth DNS DHCP TIME ELLIE Last Host — web does not shrink, gates do not open without receipts. Millennium Ladybird Servo inherit field-queen-gates-seed.json — no Chromium dependency — same doctrine when production ready.

Case study WebRTC without gatekeeper — legacy disabled WebRTC, sites broke, operator blind to STUN. Queen holds WebRTC, logs peers, archives verdicts — visibility returns.

Case study MP4 optional pack — H264 only sites failed until Chrome install. Queen mandatory in-tree closes support loop with FieldFox play confirm.

Case study sovereign SQUIDGIE during browsing — fix time before blaming site malice; navigation receipts mis-timestamp when clocks dirty.

Operator covenant clause six hold gates — Chapter 18 promise implemented in Chapter 21 tooling. Teach covenant and Queen same session — students see ethics become json verdicts.

Panel debugging queen_verdict — never treat verdict as decoration; grep panel json in CI for QUEEN_READY on release images intended for Queen posture.

Chromium family process names in seed — awareness when operators run non-FieldFox binaries; hooks apply where NEXUS installed; doctrine unchanged: hold gates, omit nothing.

nexus_bindings all true in seed — connection_gatekeeper packet_field browser_awareness honorability_db fair_ad_guardian thermal_governor sovereign_time field_dns_truth_resolver — panel should show each enabled when posture correct.

HTTP3 QUIC COOP COEP notifications push third_party_embeds mathml — each gate named in prior sections — Queen is encyclopedic surface area; operator job is hold all, not enumerate excuses to omit.

Honest rocks: seed and panel implemented; FieldFox launch implemented; queen-browser build path implemented; Ladybird production millennium; KILROY every host today field package path per Chapter 12 — QEMU bare metal honesty.

Complete gate manifest walkthrough

The seed enumerates gates in layer order: render (html_dom), script (javascript, wasm, service_workers), gpu (webgl, webgpu, canvas_2d), media (mse, eme), realtime (webrtc), network (websockets, fetch_xhr, http2_http3), storage (indexeddb, file_system_access, shared_array_buffer), sensor (geolocation, notifications, camera, microphone), device (bluetooth, usb, serial, gamepad), document (pdf_inline, svg_mathml), ui (popups, third_party_embeds), legacy (legacy_plugins_surrogate). Each id is queryable via field-queen-browser.py gate id — returns held, verdict HELD or UNHELD, layer, label. CI pipeline can loop all ids and fail build if any UNHELD.

Codec manifest walkthrough

codec_manifest returns mse_required true, mp4_mandatory true, container list including video/mp4, video codecs avc1 h264 hev1 vp9 av01, audio mp4a aac opus flac. Note string reminds MP4 H264 AAC in-tree via Gecko FFmpeg. Missing video/mp4 triggers MP4_MISSING verdict — treat as release blocker for Queen images.

field package sovereign directory layout

Queen Forge output field/sovereign/ typically bundles: KILROY kernel image with CONFIG_RTX_FIELD_DIE; initramfs or rootfs with NEXUS lib modules; FieldFox or queen-browser binary; field-services-2026 seed and data files; field-queen-gates-seed.json; sovereign-time and field-ntp keys state templates; panel TLS material; operator grep runbook markdown. Manifest sealed with Grok Build secure channel hashes — verify before bare-metal flash.

Userspace Queen and kernel kilroy_field proc interface share field die vocabulary — read both when debugging dispatch issues on sovereign hardware.

WebRTC gate deep dive

WebRTC is three surfaces: signaling over HTTPS, STUN/TURN UDP, media DTLS-SRTP. Queen holds all three in gatekeeper scoring. Packet field sentences should show signaling host, STUN server IPs, peer connection duration. Honorability consults before sustained peer egress. Operator consent gate applies to camera microphone tracks. Privacy fork that disables WebRTC entirely hides STUN from jsonl — blind operator — rejected doctrine. Lab drill: open peer connection test page, grep packet field for STUN, archive gatekeeper verdict, confirm camera prompt held not pre-granted.

MP4 MSE pipeline deep dive

MSE attaches SourceBuffers for video/mp4; Gecko media demuxes; FFmpeg decodes avc1 and mp4a; audio routed to sink; video to compositor. Mandatory in-tree means CI plays sample mp4 without network codec download. Test vectors: baseline H264 AAC, fragmented mp4, adaptive stream if enabled. Failures map to codec_manifest or missing system libs — not tell user get Chrome.

EME DRM held deep dive

EME negotiates Widevine or platform CDM; license server HTTPS flows through gatekeeper same as ads. Omitting EME does not remove Netflix-style sites from web — it sends users elsewhere. Held means license URLs visible in packet field and honorability. Fair Ad Guardian may tag license server domain separately from CDN.

Thermal Governor and thermo ethic

Per-tab GPU contexts accumulate energy proxy scores — aligns browser with ThermoAccountant ethic from engine. Abusive WebGL miner tab should spike thermal governor score before silicon throttle — operator sees software receipt before hardware pain.

Native messaging bridge protocol sketch

fieldfox-native-bridge.py receives browser events — navigation start, navigation complete, permission prompt, peer connection — forwards to connection-gatekeeper.py for intent scoring; results written to packet field jsonl. Bridge is local-only — no cloud — consistent with NEXUS local-first covenant.

Millennium Ladybird Servo migration plan

FieldFox ships 2026. Ladybird Servo inherit field-queen-gates-seed.json without Chromium — migration means copy seed, verify gate ids map to new engine APIs, run same field-queen-browser.py gate checks, play same MP4 vectors, run same WebRTC lab. No gate omission during migration — if new engine lacks WebGPU temporarily, verdict is GATES_INCOMPLETE until caught up — not shipped as Queen READY.

KILROY userspace coordination

When field package boots KILROY kernel, proc kilroy_field exposes die telemetry at syscall boundary. queen-browser or FieldFox userspace reads telemetry for HUD or panel — same data_bus vocabulary as AMOURANTHRTX Chapter 8. FCC kernel parallel Chapter 9 clamps aggressive modes from entropy feedback — browser and kernel share stability language.

Queen operator weekly checklist

field-queen-browser.py json — QUEEN_READY. gates — all_held. codecs — mp4_mandatory. Truth DNS lock in profile. sovereign time USER_OK. Play MP4 sample. WebRTC test grep STUN. Archive panel slice. Review vulnerabilities_retired browser entries with team.

Capability	Queen posture
WebGL / WebGPU	On — thermo per context
WebRTC	On — gatekeeper per peer
MSE / MP4 / H.264 / AAC	Mandatory in-tree
EME / Widevine	On — held, not omitted
Service Workers / WASM	On — entropy receipts
Geolocation / camera / mic	On — operator consent gate
Legacy plugins	WASM surrogates — surface preserved

```
NEXUS_FIELD_BROWSER_QUEEN=1 ./lib/fieldfox-launch.sh
python3 lib/field-queen-browser.py json
NEXUS_INSTALL_ROOT=Queen ./build/rtx/bin/Linux/queen-browser --sovereign --queen
```

Navigation → Packet Field → Gatekeeper → Honorability → Thermo receipt
 ↑ Truth DNS (no Google shortcut)
 ↑ Sovereign Time (SQUIDGIE witness)

Operator drills

Drill 21-A — Gate manifest. `python3 field-queen-browser.py gates` — goal: `all_held true`, total matches seed count.

Drill 21-B — MP4 in-tree. Play H.264/AAC mp4 without external browser. Goal: no optional codec pack install.

Drill 21-C — WebRTC gate. Open peer test page; grep packet field for STUN/TURN. Goal: gatekeeper verdict archived.

Drill 21-D — DNS lock. Confirm resolver 127.0.0.1 only in FieldFox profile. Goal: foreign shortcut fails.

Drill 21-E — field package audit. Inspect field/sovereign/ manifest if deployed. Goal: kernel + Queen + services documented.

Study questions

1. What is difference between hold and omit for WebRTC?
2. Why is MP4 mandatory in-tree?
3. How does queen_verdict become QUEEN_READY?
4. Name five gates from field-queen-gates-seed.json.
5. How does KILROY /proc/kilroy_field relate to Queen userspace?
6. What is FieldFox vs queen-browser ship track?
7. How does sovereign SQUIDGIE affect browser perimeter?
8. What vulnerabilities_retired entries apply to browser class?
9. Describe navigation receipt stack bottom to top.
10. Why are legacy plugins WASM surrogates instead of removed?

Chapter 22 — Glossary →

Queen gate seed held flags are boolean — CI should fail if any gate sets held false accidentally in merge conflict.

field-queen-browser.py build command refreshes panel cache before screenshot releases — stale queen_verdict misleads users.

WebGPU compute shaders in browser share thermal governor with WebGL — both bill per tab.

Service worker cache API persistence triggers storage gate receipts — long-lived caches are exfiltration risk if unheld.

SharedArrayBuffer requires COOP COEP headers — Queen holds gate with isolation policy, not disable.

HTTP/3 QUIC UDP flows appear in packet field with distinct port habits — gatekeeper learns process association over time.

Third-party iframe embeds on scholarly sites are why popups and embeds gates exist — amputation breaks citations.

WASM surrogates for Flash preserve legacy homework sites — operator schools benefit from Queen surface preservation.

fieldfox-launch.sh should be invoked from documented install root — NEXUS_INSTALL_ROOT mismatch breaks native bridge paths.

queen-browser --sovereign --queen flags link engine time and DNS posture — document in sovereign field package README.

Grok Build manifest hash verification should be step one of Queen Forge flash procedure — skip is covenant violation.

proc kilroy_field reads align with data_bus slots vocabulary — kernel userspace debug pairs Chapter 8 and Chapter 21.

browser-awareness honorability updates on tab switch — panel live view requires NEXUS daemon running.

Connection gatekeeper ten-axis intent is local inference — not cloud reputation — consistent with local-first stack.

Fair Ad Guardian reduces need for blind ad blockers that omit network surface — holds junk without amputating first-party.

EME license server flows may coincide with CDN — separate scoring axes prevent conflated verdicts.

MathML in Field Primer pages dogfoods document gates — Queen should render this chapter's equations inline.

Gamepad API gate matters for browser games — held for completeness, scored like any device API.

Web Serial and WebUSB gates protect industrial operator tools — disabling would push operators to unsafe binaries.

Queen weekly checklist should be printed beside Chapter 18 covenant poster — ethics plus json verdicts.

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Evidence anchor — grep and sources

Major claims in this chapter anchored for reproducibility. **Implemented** = grep today;
Metaphor = intuition; **Philosophy** = discipline.

Claim	Statement	Label	Evidence
QUEEN_READY	Integrated browser gates	Implemented	Build flag QUEEN_BROWSER_BUILD
WebRTC gated	Not amputated	Implemented	Through Connection Gatekeeper
Hostess 7 SDF	Segment plate doctrine	Metaphor	Not fMRI – procedural
Robot-brain stack	In-process recall	Metaphor	Architecture metaphor

```
all_gates_held = WebRTC + EME + COOP/COEP + thermo_per_context
```

```
grep -r 'QUEEN_READY\|QUEEN_BROWSER_BUILD' Navigator/
```

Source paths

- Navigator/queen-browser/
- Hostess7/scripts/field_hostess_sdf_storage.py

Chapter summary — before you turn the page

Queen holds all gates in-engine — WebRTC gated, MP4 in-tree, rtx spine shared. Sovereign browser is perimeter, not amputation.

Glossary — Field Technology v5

Master rocks and open questions

Consolidated honesty table — every major claim that marketing might exaggerate, with current evidence level. Supersedes scattered rocks in Chapter 12 and the site index. Update this table when implementation catches aspiration.

Claim	Operator reality	Label	Evidence / gap
This is a certified textbook	Long-form operator course; manuscript not publisher-certified	Philosophy	Edition subtitle, not external validation
Textbook of 2026	Edition name and teaching ambition	Philosophy	Not third-party curriculum approval
Field literacy as greatest weapon	Read/write continuous state locally; distrust single dashboards	Philosophy	Operator discipline, not ordnance
Living thermodynamic computer	ThermoAccountant in stderr each dispatch	Implemented	grep THERMO
Landauer joules from GPU	Proxy integral entropyThisFrame	Metaphor	Not package wattmeter
Packet field sees everything	Local sockets + heuristics on your machine	Implemented	Not internet omniscience
RF planetary shell	planetary_weave.comp visual stack	Visual	Not spectrum analyzer
Queen holds all gates	WebRTC/EME through gatekeeper when built	Implemented	QUEEN_READY required
DARPA-grade / robot brain	In-process Queen architecture metaphor for gate+recall stack	Metaphor	Not a government program deliverable
Hostess 7 SDF brain imaging	Procedural SDF plates + segment doctrine	Metaphor	Not fMRI; Hostess7/ workflow

Claim	Operator reality	Label	Evidence / gap
KILROY on daily driver host	Kernel image exists; dev may use generic Linux	Implemented	QEMU/bare-metal vs laptop
Love / God chapters prove physics	Sacred operator language beside math	Philosophy	Ch 16–18 optional track
Sub-micron SEM in shaders	Procedural pixel detail tiers	Metaphor	Spiderweb simulate tier
Tesla valve in chassis	TESLA_R_* bias; data_bus slots	Metaphor	No fluidic part shipped
96% lies / device cannot drift	Security posture narrative	Philosophy	Not statistical claim
Illustration density ~1/1000 words	Mayer segmenting standard for v5	Philosophy	Editorial rule, Ch 1
Shannon H on fabric texels	Entropy Oracle on <i>files</i>	Implemented	Separate from ThermoAccountant
Sealed / sovereign time	TotalTime::seal() + HMAC pulses	Implemented	SQUIDGIE on verify fail
x86.comp decorative	Default ./linux.sh run path	Implemented	Field Die product
Equations without bindings	Theory vocabulary until grep proves	Metaphor	Ch 12 capstone table layers

Open questions (honest gaps — not roadmap promises): calibrated GPU joules vs proxy entropy; wire DNSSEC validation beyond stub counters; Queen production parity across all gate classes; reproducible cross-host sovereign time under adversarial NTP. File issues against AMOURANTHRTX / NEXUS when you close a gap — then update this row.

Glossary — Field Technology v5

This chapter is an encyclopedic glossary for the full textbook. Every term used across Chapters 1–21 appears here in prose *and* in the master table at the end. When a term has both poetry and code meanings, both are named. Read alphabetically by section; grep the table when you are in a hurry.

Figure 22.1 — Overloaded words disambiguated: FieldSocket ≠ packet field ≠ RF shell.

On the way — what you will learn

The glossary is the encyclopedic index of Field Technology v5 vocabulary — formal definitions beside honesty labels. On the way you will disambiguate FieldSocket from packet field, RF meanings, entropy layers, and every rock-named knob.

On the way — Glossary

- Alphabetical term lookup
- Cross-links to chapter deep dives
- Disambiguation tables for overloaded words
- Keep open while reading any chapter

IN THIS CHAPTER	
1. Glossary — Field Technology v5	7. F
2. A	8. G
3. B	9. H
4. C	10. I
5. D	11. K
6. E	12. L

13. M

14. N

Status labels from Chapter 1 still apply: **Implemented** means grep-able source; **Metaphor** means intuition not SI units; **Philosophy** means operator discipline.

A

Adaptive scale

Adaptive scale — Resolution scaling 320×200 toward 4K+ — honest performance trade.

In textbook usage, *Adaptive scale* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Adaptive scale in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Adept tier

Adept tier — Spiderweb level with SimulateSubMicron and target clock control.

In textbook usage, *Adept tier* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Adept tier in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

AmmoOS

AmmoOS — Guest operating environment living inside Field Die address space — pumped by x86.comp each frame.

In textbook usage, *AmmoOS* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet AmmoOS in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Amouranth

Amouranth — Engine namesake spirit — courage to be seen while holding boundaries.

In textbook usage, *Amouranth* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Amouranth in a panel slice, jsonl receipt, or shader header, this definition

is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

AMOURANTHRTX

AMOURANTHRTX — The Vulkan field engine — fabric channels, Field Die, dispatch loop, ELLIE logging. Default `/linux.sh` run enters Field Die path.

In textbook usage, *AMOURANTHRTX* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet AMOURANTHRTX in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

AnalogFields

AnalogFields — Host-side knobs for Phi, Thermo, Flow coupling — GateFidelity and related parameters the prompt terminal can set.

In textbook usage, *AnalogFields* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet AnalogFields in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Apocalypse handler

Apocalypse handler — ELLIE fail-closed path when session entropy verification fails — engine abort posture mirrored in perimeter grep discipline.

In textbook usage, *Apocalypse handler* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Apocalypse handler in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

B

Big Grin HUD

Big Grin HUD — 172×48 character HUD rendered by `x86.comp` inside Field Die — operator-facing die-resident display.

In textbook usage, *Big Grin HUD* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Big Grin HUD in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

browser-awareness

browser-awareness — NEXUS module reporting active tab honorability — feeds Queen panel slice.

In textbook usage, *browser-awareness* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet browser-awareness in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

C

CANVAS.comp

CANVAS.comp — Classic canvas compute shader — thermo and RT demos on PushConstants path.

In textbook usage, *CANVAS.comp* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet CANVAS.comp in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Captain Ellie

Captain Ellie — Persona encoding ELLIE.hpp covenant — TotalTime seal, verify, THERMO logging categories.

In textbook usage, *Captain Ellie* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Captain Ellie in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

CC BY-NC-SA

CC BY-NC-SA — Field Primer license — teach freely, name rocks, noncommercial share-alike.

In textbook usage, *CC BY-NC-SA* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet CC BY-NC-SA in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

CFL

CFL — Courant–Friedrichs–Lewy stability condition — host refuses fabric steps that would outrun the mesh.

In textbook usage, *CFL* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet CFL in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Connection Gatekeeper

Connection Gatekeeper — NEXUS 10-axis intent scorer — every connection earns verdict before sustained egress.

In textbook usage, *Connection Gatekeeper* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Connection Gatekeeper in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

COOP/COEP

COOP/COEP — Cross-origin isolation headers — SharedArrayBuffer gate held with policy.

In textbook usage, *COOP/COEP* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet COOP/COEP in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

D

data_bus

data_bus — Sixty-four word telemetry spine per dispatch — FCC floats, thermo mirrors, input, Tesla bias.

In textbook usage, *data_bus* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *data_bus* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

dig +trace

dig +trace — DNS trace from root — Truth Resolver discipline, no shortcut.

In textbook usage, *dig +trace* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *dig +trace* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

DNS threat guard

DNS threat guard — Rate limits and permanent blocks for abusive DNS query patterns — retired dig fork bomb class.

In textbook usage, *DNS threat guard* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *DNS threat guard* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

dns-egress-integrity

dns-egress-integrity — Hashes DNS answers so tamper in flight is visible to operator.

In textbook usage, *dns-egress-integrity* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *dns-egress-integrity* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

dns-service-takeover

dns-service-takeover — Waits for healthy loopback DNS before steering resolv.conf to Truth Resolver.

In textbook usage, *dns-service-takeover* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *dns-service-takeover* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

E

ELLIE

ELLIE — Logging and covenant framework in engine — MAIN, VULKAN, CANVAS, THERMO, STATUS, RTXPROBE categories.

Figure 22.3 — E entries: entropy layers never summed without labels.

In textbook usage, *ELLIE* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *ELLIE* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

ellie-last-host

ellie-last-host — Python module for sole-survivor DNS/DHCP/TIME posture when NEXUS_LAST_HOST=1.

In textbook usage, *ellie-last-host* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *ellie-last-host* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

EME

EME — Encrypted Media Extensions — Widevine held in Queen, not omitted.

In textbook usage, *EME* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *EME* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Energy can be moved

Energy can be moved — Axiom — coupling moves irreversibility; accounting stays honest.

In textbook usage, *Energy can be moved* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Energy can be moved in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Entropy Oracle

Entropy Oracle — NEXUS Shannon H file analyzer — surprise measurement on local files.

In textbook usage, *Entropy Oracle* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Entropy Oracle in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

entropyThisFrame

entropyThisFrame — ThermoAccountant per-dispatch entropy increment — receipt time ran forward in fabric.

In textbook usage, *entropyThisFrame* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet entropyThisFrame in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

entropy_tag

entropy_tag — 12-hex tag on sovereign pulses — hashes mono_ns and pulse number.

In textbook usage, *entropy_tag* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet entropy_tag in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

F

Fair Ad Guardian

Fair Ad Guardian — First-party vs junk ad scoring — Queen browser egress layer.

Figure 22.2 — F entries: FieldSocket, Field Die — die-resident vocabulary.

Claim: Default ./linux.sh run boots Field Die — x86.comp path

In textbook usage, *Fair Ad Guardian* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Fair Ad Guardian in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

FCC

FCC — Field Control Coupling floats in data_bus slots 16–23 — stability knobs.

In textbook usage, *FCC* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet FCC in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Field DHCP

Field DHCP — NEXUS DHCP v3 — LAN bind, option-50 match, ping conflict, DNS option to Truth Resolver.

In textbook usage, *Field DHCP* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Field DHCP in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Field Die

Field Die — GPU SSBO simulating x86 guest with 64 MiB RAM — addressable universe on silicon.

In textbook usage, *Field Die* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Field Die in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

field jsonl

field jsonl — Append-only packet field log — grep forensic backbone.

In textbook usage, *field jsonl* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet field jsonl in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Field layer

Field layer — Composable L0–L9 die layers — RAM, VGA, FAT, audio, BIOS pumped via FieldLayer::pumpAll.

In textbook usage, *Field layer* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Field layer in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Field literacy

Field literacy — Reading continuous state, imposing boundary conditions — greatest operator weapon.

In textbook usage, *Field literacy* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Field literacy in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

field package

field package — Sealed sovereign manifest under field/sovereign/ — kernel, Queen, services, grep discipline.

In textbook usage, *field package* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet field package in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

field-ntp-2026

field-ntp-2026 — NTP mode-4 server gated on sovereign pulse health — UDP 123.

In textbook usage, *field-ntp-2026* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *field-ntp-2026* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

field-services-2026

field-services-2026 — Unified 2026 public services panel manifest — DNS, DHCP, NTP, sovereign slices.

In textbook usage, *field-services-2026* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *field-services-2026* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

FieldFox

FieldFox — Hardened Gecko fork — telemetry stripped, Truth DNS, full codec tree, NEXUS native bridge.

In textbook usage, *FieldFox* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *FieldFox* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

FieldSocket

FieldSocket — Push constants block for x86 canvas dispatch — carries sealed_time among die parameters.

In textbook usage, *FieldSocket* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *FieldSocket* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

FieldWebPanel

FieldWebPanel — In-engine web panel host inside queen-browser for Field Primer and operator UI.

In textbook usage, *FieldWebPanel* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *FieldWebPanel* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

FIELD_LAYOUT_VERSION

FIELD_LAYOUT_VERSION — Descriptor layout version 5 — host and shader must match.

In textbook usage, *FIELD_LAYOUT_VERSION* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *FIELD_LAYOUT_VERSION* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Flow

Flow — Momentum and gradient fabric channel — binding 10, .gb vector components per texel.

In textbook usage, *Flow* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *Flow* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

freq_sum_khz

freq_sum_khz — Sum of per-CPU sysfs frequencies in pulse — squidgie detection input.

In textbook usage, *freq_sum_khz* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *freq_sum_khz* in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

G

Gate fidelity

Gate fidelity — Analog knob affecting Phi gate potential coupling — prompt terminal settable.

In textbook usage, *Gate fidelity* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet *Gate fidelity* in a panel slice, jsonl receipt, or shader header, this

definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Gatekeeper

Gatekeeper — Shorthand for Connection Gatekeeper — connection scoring to verdict.

In textbook usage, *Gatekeeper* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Gatekeeper in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

GPS precision

GPS precision — Sub-micron ENU node tooling — requires stable UTC labels from sovereign time.

In textbook usage, *GPS precision* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet GPS precision in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Grep discipline

Grep discipline — Operator ethic — truth is what survives grep, not screenshots.

In textbook usage, *Grep discipline* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Grep discipline in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Grok

Grok — Co-documentation contributor — light beside text, not operator conscience.

In textbook usage, *Grok* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Grok in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Grok Build

Grok Build — Secure build channel referenced in sovereign field package manifests.

In textbook usage, *Grok Build* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Grok Build in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

H

hardwareFabric

hardwareFabric — Read-only Spiderweb mirror of averaged fabric channels — dashboard not second sim.

In textbook usage, *hardwareFabric* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet hardwareFabric in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Holographic boundary

Holographic boundary — Philosophy chapter 17 motif — God/Truth at die edge, not calorimetry substitute.

In textbook usage, *Holographic boundary* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Holographic boundary in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Honest rocks

Honest rocks — Chapter 12 table — claims with Implemented/Metaphor/Philosophy labels.

In textbook usage, *Honest rocks* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Honest rocks in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Honorability

Honorability — Site trust database consulted before browser egress proceeds.

In textbook usage, *Honorability* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Honorability in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Honorability DB

Honorability DB — Persistent site trust store — browser-awareness and gatekeeper consult it.

In textbook usage, *Honorability DB* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Honorability DB in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Hostess 7

Hostess 7 — Watchguard persona in Queen chrome deck — Forever Watchguard split UI motif.

In textbook usage, *Hostess 7* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Hostess 7 in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Hostess7

Hostess7 — DNS admin persona — production passkey via env not seed.

In textbook usage, *Hostess7* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Hostess7 in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

HTTP/3 QUIC

HTTP/3 QUIC — Modern transport gate held — scored like HTTP/2 flows.

In textbook usage, *HTTP/3 QUIC* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet HTTP/3 QUIC in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

I

Implemented tag

Implemented tag — Honesty label for grep-able shipped code.

In textbook usage, *Implemented tag* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Implemented tag in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

IndexedDB

IndexedDB — Browser storage gate held in Queen — receipts required for persistence.

In textbook usage, *IndexedDB* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet IndexedDB in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Intent scoring

Intent scoring — Gatekeeper axis evaluation of connection purpose vs process path habit.

In textbook usage, *Intent scoring* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Intent scoring in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

K

KILROY

KILROY — Field OS kernel — CONFIG_RTX_FIELD_DIE, /proc/kilroy_field syscall boundary, FCC parallel.

In textbook usage, *KILROY* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet KILROY in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

kilroy_field

kilroy_field — Procs interface exposing field die telemetry at kernel boundary for sovereign deployments.

In textbook usage, *kilroy_field* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet kilroy_field in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.



Ladybird

Ladybird — Millennium independent browser engine track — inherits Queen gate seed.

In textbook usage, *Ladybird* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Ladybird in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Landauer bound

Landauer bound — Minimum thermodynamic cost per bit erase — ThermoAccountant proxy honors discipline.

In textbook usage, *Landauer bound* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Landauer bound in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Legacy plugins

Legacy plugins — Flash/Java/Silverlight — WASM surrogates in Queen, not native VMs.

In textbook usage, *Legacy plugins* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Legacy plugins in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

LOG_THERMO

LOG_THERMO — ELLIE grep category for per-frame entropy and maintenance lines.

In textbook usage, *LOG_THERMO* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet LOG_THERMO in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Loopback-first

Loopback-first — Default bind 127.0.0.1 for DNS and sovereign time — widen explicitly.

In textbook usage, *Loopback-first* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Loopback-first in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Love coupling

Love coupling — Chapter 16 sacred motif — coupled evolution with consent, not telemetry extraction.

In textbook usage, *Love coupling* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Love coupling in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

M

Maintenance cost

Maintenance cost — Thermo price of frame coherence — comparative proxy, not utility bill.

In textbook usage, *Maintenance cost* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Maintenance cost in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Maxwell coupling

Maxwell coupling — Phi fabric evolution tied to creditor Maxwell — wave channel physics metaphor.

In textbook usage, *Maxwell coupling* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Maxwell coupling in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Metaphor tag

Metaphor tag — Honesty label for poetic knobs not yet SI-calibrated.

In textbook usage, *Metaphor tag* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Metaphor tag in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Micron witness

Micron witness — SHA-256 fingerprint binding monotonic instant to sysfs CPU frequencies in sovereign pulses.

In textbook usage, *Micron witness* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Micron witness in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

MP4

MP4 — Mandatory container in Queen — H.264/AAC in-tree via Gecko media and FFmpeg.

In textbook usage, *MP4* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet MP4 in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

MSE

MSE — Media Source Extensions — mandatory for Queen MP4 path.

In textbook usage, *MSE* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet MSE in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

N

Native messaging

Native messaging — FieldFox bridge to NEXUS gatekeeper from browser extension channel.

In textbook usage, *Native messaging* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Native messaging in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

NEXUS

NEXUS — Local-first defense stack — packet field, panel :9477, DNS, DHCP, time, Queen slices.

In textbook usage, *NEXUS* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet NEXUS in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

NEXUS-Shield

NEXUS-Shield — Distribution and daemon wrapping NEXUS lib modules — `nexus.sh` entry.

In textbook usage, *NEXUS-Shield* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet NEXUS-Shield in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

NEXUS_FIELD_BROWSER_QUEEN

NEXUS_FIELD_BROWSER_QUEEN — Enable Queen browser posture — hold all gates.

In textbook usage, *NEXUS_FIELD_BROWSER_QUEEN* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet NEXUS_FIELD_BROWSER_QUEEN in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

NEXUS_FIELD_SERVICES_PUBLIC

NEXUS_FIELD_SERVICES_PUBLIC — Explicit WAN exposure opt-in for DNS, DHCP, NTP — never default.

In textbook usage, *NEXUS_FIELD_SERVICES_PUBLIC* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet NEXUS_FIELD_SERVICES_PUBLIC in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

NEXUS_LAST_HOST

NEXUS_LAST_HOST — Environment flag for ELLIE sole survivor — global services on last machine.

In textbook usage, *NEXUS_LAST_HOST* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet NEXUS_LAST_HOST in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Nick

Nick — Builder beside stack — honest fields, no phone-home.

In textbook usage, *Nick* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Nick in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

O

Operator

Operator — Human at keyboard — final authority; daemons assist, do not inherit conscience.

In textbook usage, *Operator* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Operator in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Operator covenant

Operator covenant — Chapter 18 clauses — teach, build local, honor creditors, hold gates.

In textbook usage, *Operator covenant* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Operator covenant in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

P

Packet field

Packet field — NEXUS jsonl telemetry turning sockets into sentences — TX/RX, ports, paths, verdicts.

In textbook usage, *Packet field* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Packet field in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Phi

Phi — Wave and gate potential fabric channel — binding 8.

In textbook usage, *Phi* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Phi in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Philosophy tag

Philosophy tag — Honesty label for operator discipline not a sensor reading.

In textbook usage, *Philosophy tag* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Philosophy tag in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Pool NTP

Pool NTP — Public NTP pools — not sole authority under sovereign-first policy.

In textbook usage, *Pool NTP* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Pool NTP in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Primary phase

Primary phase — DNS takeover phase when loopback resolver healthy — nft blocks foreign egress.

In textbook usage, *Primary phase* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Primary phase in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Probe injection

Probe injection — Mouse/input injection dissipation counted in thermo receipts.

In textbook usage, *Probe injection* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Probe injection in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Propalactic

Propalactic — Large-scale Phi forcing scale — cosmic knob labeled metaphor unless calibrated.

In textbook usage, *Propalactic* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Propalactic in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Puny tier

Puny tier — Spiderweb mastery level — sysfs clocks, status log.

In textbook usage, *Puny tier* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Puny tier in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Push constants

Push constants — Vulkan small uniform block per dispatch — FieldSocket on x86 path.

In textbook usage, *Push constants* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Push constants in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Q

Queen

Queen — Field Browser Queen doctrine — all capabilities present, all gates held.

Figure 22.5 — Q entries: Queen holds all gates.

Claim: QUEEN_READY — all gates held in-engine

In textbook usage, *Queen* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Queen in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Queen chrome

Queen chrome — Aqua/rose 2026 aesthetic — QueenBoot.comp and RTX Zero panel rhyme.

In textbook usage, *Queen chrome* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Queen chrome in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Queen Forge

Queen Forge — Packaging pipeline producing sovereign field bundles for KILROY deployments.

In textbook usage, *Queen Forge* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Queen Forge in a panel slice, jsonl receipt, or shader header, this

definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

queen-browser

queen-browser — AMOURANTHRTX Vulkan+SDL3 in-engine browser binary — sovereign RTX field path.

In textbook usage, *queen-browser* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet queen-browser in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

QueenBoot.comp

QueenBoot.comp — Shader chrome boot path for in-engine queen-browser aqua/rose aesthetic.

In textbook usage, *QueenBoot.comp* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet QueenBoot.comp in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

QUEEN_READY

QUEEN_READY — field-queen-browser.py verdict when gates complete and MP4 mandatory satisfied.

In textbook usage, *QUEEN_READY* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet QUEEN_READY in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

R

Reality is 3D

Reality is 3D — Axiom — state occupies addressable space: texels, die bytes, sockets.

Figure 22.4 — R entries: three RF meanings disambiguated.

In textbook usage, *Reality is 3D* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Reality is 3D in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

RTX Zero

RTX Zero — NEXUS panel mode — Aqua chrome, cache-first refresh via ?rtx=1.

In textbook usage, *RTX Zero* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet RTX Zero in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

RTX-DOS

RTX-DOS — Guest DOS environment in Field Die — pumped x86 path default run mode.

In textbook usage, *RTX-DOS* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet RTX-DOS in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

RTXProbe

RTXProbe — Optional GPU timestamp probes — RTX_PROBES=1, zero cost when off.

In textbook usage, *RTXProbe* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet RTXProbe in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

S

Seal forward

Seal forward — Monotonic genesis and pulse chains only advance — never rewrite physics time.

In textbook usage, *Seal forward* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Seal forward in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Sealed time

Sealed time — Monotonic session clock in FieldSocket — immune to frame-rate drift.

In textbook usage, *Sealed time* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Sealed time in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Servo

Servo — Mozilla layout engine — millennium track beside Ladybird.

In textbook usage, *Servo* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Servo in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Shannon H

Shannon H — Entropy Oracle surprise measure — creditor Shannon honored in NEXUS tooling.

In textbook usage, *Shannon H* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Shannon H in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Sovereign time

Sovereign time — Operator-owned signed pulses on UDP 9123 — sovereign-time.py.

In textbook usage, *Sovereign time* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Sovereign time in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Spiderweb

Spiderweb — hardwareFabric graph mirroring fabric averages — Puny/Adept/Tidewalker tiers.

In textbook usage, *Spiderweb* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Spiderweb in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

SQUIDGIE

SQUIDGIE — Sovereign time tamper verdict — clocks disagree at receive; grep it.

In textbook usage, *SQUIDGIE* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet SQUIDGIE in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Stratum

Stratum — NTP hierarchy level — must trace to signed sovereign pulses in 2026 stack.

In textbook usage, *Stratum* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Stratum in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Sub-micron

Sub-micron — Spiderweb Adept simulation tier — procedural detail, not lab SEM promise.

In textbook usage, *Sub-micron* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Sub-micron in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

T

Terror-threat posture

Terror-threat posture — Assume adversary knows field literacy — verify at receive, loopback-first.

In textbook usage, *Terror-threat posture* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Terror-threat posture in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Tesla valve

Tesla valve — Directional flow resistance bias in fabric — forward ease, reverse resist.

In textbook usage, *Tesla valve* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Tesla valve in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Thermal Governor

Thermal Governor — Per-tab energy honesty layer in Queen browser stack.

In textbook usage, *Thermal Governor* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Thermal Governor in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Thermo

Thermo — Heat and entropy density fabric channel — binding 9.

In textbook usage, *Thermo* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Thermo in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Thermo boundary

Thermo boundary — Maintains coherence with previous frame — prevMaintCost in receipts.

In textbook usage, *Thermo boundary* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Thermo boundary in a panel slice, jsonl receipt, or shader header,

this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

ThermoAccountant

ThermoAccountant — Per-frame entropy ledger SSBO binding 2 — entropyThisFrame receipt.

In textbook usage, *ThermoAccountant* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet ThermoAccountant in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Threat panel

Threat panel — <https://127.0.0.1:9477/> — command, packets, threats, DNS, library, system.

In textbook usage, *Threat panel* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Threat panel in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Tidewalker tier

Tidewalker tier — Full spiderweb graph override — highest operator mirror control.

In textbook usage, *Tidewalker tier* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Tidewalker tier in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Time is linear

Time is linear — Axiom — logs are timeline; sealed clocks do not rewrite physics time.

In textbook usage, *Time is linear* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Time is linear in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

TotalTime

TotalTime — ELLIE session clock API — seal and verify forward-only genesis.

In textbook usage, *TotalTime* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet TotalTime in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Truth DNS

Truth DNS — Loopback-first resolver — RFC trace, no foreign shortcut, field-dns.py.

In textbook usage, *Truth DNS* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Truth DNS in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

U

USER_OK

USER_OK — Sovereign verify clean verdict — opposite of SQUIDGIE.

In textbook usage, *USER_OK* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet USER_OK in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

V

Verify at receive

Verify at receive — Receiver double-check ethic — DNS hash, DHCP conflict, time SQUIDGIE.

In textbook usage, *Verify at receive* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Verify at receive in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

VGA

VGA — Text mode buffer at guest 0xB8000 inside Field Die.

In textbook usage, *VGA* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet VGA in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Vulkan fabric

Vulkan fabric — `rtx()` singleton — device, queues, hardwareFabric, VRAM budget.

In textbook usage, *Vulkan fabric* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Vulkan fabric in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.



WASM surrogates

WASM surrogates — Replacement layer for dead plugins — preserves surface without JVM/Flash VM.

In textbook usage, *WASM surrogates* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet WASM surrogates in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

WebRTC

WebRTC — Realtime peer API held in Queen — gatekeeper per peer, not disabled.

In textbook usage, *WebRTC* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet WebRTC in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Widevine

Widevine — DRM system held via EME — license flows scored, not amputated.

In textbook usage, *Widevine* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Widevine in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

X

x86.comp

x86.comp — Field Die compute shader — guest interpretation and Big Grin HUD.

In textbook usage, *x86.comp* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet x86.comp in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Z

Zachary Geurts

Zachary Geurts — Architect and author — AMOURANTHRTX, NEXUS-Shield, Queen, Field Primer.

In textbook usage, *Zachary Geurts* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet Zachary Geurts in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

ZMM1024

ZMM1024 — Tile cache tail in FieldX86Die SSBO — shader-side fabric sample cache for HUD.

In textbook usage, *ZMM1024* appears wherever operators cross engine, NEXUS, and Queen boundaries. If you meet ZMM1024 in a panel slice, jsonl receipt, or shader header, this definition is the canonical Field Technology v5 reading — not generic industry jargon unless explicitly noted.

Master summary table

Quick reference — same terms, one line each.

Term	Definition
Adaptive scale	Resolution scaling 320×200 toward 4K+ — honest performance trade.
Adept tier	Spiderweb level with SimulateSubMicron and target clock control.
AmmoOS	Guest operating environment living inside Field Die address space — pumped by x86.comp each frame.
Amouranth	Engine namesake spirit — courage to be seen while holding boundaries.
AMOURANTHRTX	The Vulkan field engine — fabric channels, Field Die, dispatch loop, ELLIE logging. Default <code>./linux.sh</code> run enters Field Die path.
AnalogFields	Host-side knobs for Phi, Thermo, Flow coupling — GateFidelity and related parameters the prompt terminal can set.
Apocalypse handler	ELLIE fail-closed path when session entropy verification fails — engine abort posture mirrored in perimeter grep discipline.
Big Grin HUD	172×48 character HUD rendered by x86.comp inside Field Die — operator-facing die-resident display.
browser-awareness	NEXUS module reporting active tab honorability — feeds Queen panel slice.
CANVAS.comp	Classic canvas compute shader — thermo and RT demos on PushConstants path.
Captain Ellie	Persona encoding ELLIE.hpp covenant — TotalTime seal, verify, THERMO logging

Term	Definition
	categories.
CC BY-NC-SA	Field Primer license — teach freely, name rocks, noncommercial share-alike.
CFL	Courant–Friedrichs–Lewy stability condition — host refuses fabric steps that would outrun the mesh.
Connection Gatekeeper	NEXUS 10-axis intent scorer — every connection earns verdict before sustained egress.
COOP/COEP	Cross-origin isolation headers — SharedArrayBuffer gate held with policy.
data_bus	Sixty-four word telemetry spine per dispatch — FCC floats, thermo mirrors, input, Tesla bias.
dig +trace	DNS trace from root — Truth Resolver discipline, no shortcut.
DNS threat guard	Rate limits and permanent blocks for abusive DNS query patterns — retired dig fork bomb class.
dns-egress-integrity	Hashes DNS answers so tamper in flight is visible to operator.
dns-service-takeover	Waits for healthy loopback DNS before steering resolv.conf to Truth Resolver.
ELLIE	Logging and covenant framework in engine — MAIN, VULKAN, CANVAS, THERMO, STATUS, RTXPROBE categories.
ellie-last-host	Python module for sole-survivor DNS/DHCP/TIME posture when NEXUS_LAST_HOST=1.

Term	Definition
EME	Encrypted Media Extensions — Widevine held in Queen, not omitted.
Energy can be moved	Axiom — coupling moves irreversibility; accounting stays honest.
Entropy Oracle	NEXUS Shannon H file analyzer — surprise measurement on local files.
entropyThisFrame	ThermoAccountant per-dispatch entropy increment — receipt time ran forward in fabric.
entropy_tag	12-hex tag on sovereign pulses — hashes mono_ns and pulse number.
Fair Ad Guardian	First-party vs junk ad scoring — Queen browser egress layer.
FCC	Field Control Coupling floats in data_bus slots 16–23 — stability knobs.
Field DHCP	NEXUS DHCP v3 — LAN bind, option-50 match, ping conflict, DNS option to Truth Resolver.
Field Die	GPU SSBO simulating x86 guest with 64 MiB RAM — addressable universe on silicon.
field jsonl	Append-only packet field log — grep forensic backbone.
Field layer	Composable L0–L9 die layers — RAM, VGA, FAT, audio, BIOS pumped via FieldLayer::pumpAll.
Field literacy	Reading continuous state, imposing boundary conditions — greatest operator weapon.

Term	Definition
field package	Sealed sovereign manifest under field/sovereign/ — kernel, Queen, services, grep discipline.
field-ntp-2026	NTP mode-4 server gated on sovereign pulse health — UDP 123.
field-services-2026	Unified 2026 public services panel manifest — DNS, DHCP, NTP, sovereign slices.
FieldFox	Hardened Gecko fork — telemetry stripped, Truth DNS, full codec tree, NEXUS native bridge.
FieldSocket	Push constants block for x86 canvas dispatch — carries sealed_time among die parameters.
FieldWebPanel	In-engine web panel host inside queen-browser for Field Primer and operator UI.
FIELD_LAYOUT_VERSION	Descriptor layout version 5 — host and shader must match.
Flow	Momentum and gradient fabric channel — binding 10, .gb vector components per texel.
freq_sum_khz	Sum of per-CPU sysfs frequencies in pulse — squidgie detection input.
Gate fidelity	Analog knob affecting Phi gate potential coupling — prompt terminal settable.
Gatekeeper	Shorthand for Connection Gatekeeper — connection scoring to verdict.
GPS precision	Sub-micron ENU node tooling — requires stable UTC labels from sovereign time.

Term	Definition
Grep discipline	Operator ethic — truth is what survives grep, not screenshots.
Grok	Co-documentation contributor — light beside text, not operator conscience.
Grok Build	Secure build channel referenced in sovereign field package manifests.
hardwareFabric	Read-only Spiderweb mirror of averaged fabric channels — dashboard not second sim.
Holographic boundary	Philosophy chapter 17 motif — God/Truth at die edge, not calorimetry substitute.
Honest rocks	Chapter 12 table — claims with Implemented/Metaphor/Philosophy labels.
Honorability	Site trust database consulted before browser egress proceeds.
Honorability DB	Persistent site trust store — browser-awareness and gatekeeper consult it.
Hostess 7	Watchguard persona in Queen chrome deck — Forever Watchguard split UI motif.
Hostess7	DNS admin persona — production passkey via env not seed.
HTTP/3 QUIC	Modern transport gate held — scored like HTTP/2 flows.
Implemented tag	Honesty label for grep-able shipped code.

Term	Definition
IndexedDB	Browser storage gate held in Queen — receipts required for persistence.
Intent scoring	Gatekeeper axis evaluation of connection purpose vs process path habit.
KILROY	Field OS kernel — CONFIG_RTX_FIELD_DIE, /proc/kilroy_field syscall boundary, FCC parallel.
kilroy_field	Procs interface exposing field die telemetry at kernel boundary for sovereign deployments.
Ladybird	Millennium independent browser engine track — inherits Queen gate seed.
Landauer bound	Minimum thermodynamic cost per bit erase — ThermoAccountant proxy honors discipline.
Legacy plugins	Flash/Java/Silverlight — WASM surrogates in Queen, not native VMs.
LOG_THERMO	ELLIE grep category for per-frame entropy and maintenance lines.
Loopback-first	Default bind 127.0.0.1 for DNS and sovereign time — widen explicitly.
Love coupling	Chapter 16 sacred motif — coupled evolution with consent, not telemetry extraction.
Maintenance cost	Thermo price of frame coherence — comparative proxy, not utility bill.
Maxwell coupling	Phi fabric evolution tied to creditor Maxwell — wave channel physics metaphor.

Term	Definition
Metaphor tag	Honesty label for poetic knobs not yet SI-calibrated.
Micron witness	SHA-256 fingerprint binding monotonic instant to sysfs CPU frequencies in sovereign pulses.
MP4	Mandatory container in Queen — H.264/AAC in-tree via Gecko media and FFmpeg.
MSE	Media Source Extensions — mandatory for Queen MP4 path.
Native messaging	FieldFox bridge to NEXUS gatekeeper from browser extension channel.
NEXUS	Local-first defense stack — packet field, panel :9477, DNS, DHCP, time, Queen slices.
NEXUS-Shield	Distribution and daemon wrapping NEXUS lib modules — nexus.sh entry.
NEXUS_FIELD_BROWSER_QUEEN	Enable Queen browser posture — hold all gates.
NEXUS_FIELD_SERVICES_PUBLIC	Explicit WAN exposure opt-in for DNS, DHCP, NTP — never default.
NEXUS_LAST_HOST	Environment flag for ELLIE sole survivor — global services on last machine.
Nick	Builder beside stack — honest fields, no phone-home.
Operator	Human at keyboard — final authority; daemons assist, do not inherit conscience.

Term	Definition
Operator covenant	Chapter 18 clauses — teach, build local, honor creditors, hold gates.
Packet field	NEXUS jsonl telemetry turning sockets into sentences — TX/RX, ports, paths, verdicts.
Phi	Wave and gate potential fabric channel — binding 8.
Philosophy tag	Honesty label for operator discipline not a sensor reading.
Pool NTP	Public NTP pools — not sole authority under sovereign-first policy.
Primary phase	DNS takeover phase when loopback resolver healthy — nft blocks foreign egress.
Probe injection	Mouse/input injection dissipation counted in thermo receipts.
Propalactic	Large-scale Phi forcing scale — cosmic knob labeled metaphor unless calibrated.
Puny tier	Spiderweb mastery level — sysfs clocks, status log.
Push constants	Vulkan small uniform block per dispatch — FieldSocket on x86 path.
Queen	Field Browser Queen doctrine — all capabilities present, all gates held.
Queen chrome	Aqua/rose 2026 aesthetic — QueenBoot.comp and RTX Zero panel rhyme.

Term	Definition
Queen Forge	Packaging pipeline producing sovereign field bundles for KILROY deployments.
queen-browser	AMOURANTHRTX Vulkan+SDL3 in-engine browser binary — sovereign RTX field path.
QueenBoot.comp	Shader chrome boot path for in-engine queen-browser aqua/rose aesthetic.
QUEEN_READY	field-queen-browser.py verdict when gates complete and MP4 mandatory satisfied.
Reality is 3D	Axiom — state occupies addressable space: texels, die bytes, sockets.
RTX Zero	NEXUS panel mode — Aqua chrome, cache-first refresh via ?rtx=1.
RTX-DOS	Guest DOS environment in Field Die — pumped x86 path default run mode.
RTXProbe	Optional GPU timestamp probes — RTX_PROBES=1, zero cost when off.
Seal forward	Monotonic genesis and pulse chains only advance — never rewrite physics time.
Sealed time	Monotonic session clock in FieldSocket — immune to frame-rate drift.
Servo	Mozilla layout engine — millennium track beside Ladybird.
Shannon H	Entropy Oracle surprise measure — creditor Shannon honored in NEXUS tooling.

Term	Definition
Sovereign time	Operator-owned signed pulses on UDP 9123 — sovereign-time.py.
Spiderweb	hardwareFabric graph mirroring fabric averages — Puny/Adept/Tidewalker tiers.
SQUIDGIE	Sovereign time tamper verdict — clocks disagree at receive; grep it.
Stratum	NTP hierarchy level — must trace to signed sovereign pulses in 2026 stack.
Sub-micron	Spiderweb Adept simulation tier — procedural detail, not lab SEM promise.
Terror-threat posture	Assume adversary knows field literacy — verify at receive, loopback-first.
Tesla valve	Directional flow resistance bias in fabric — forward ease, reverse resist.
Thermal Governor	Per-tab energy honesty layer in Queen browser stack.
Thermo	Heat and entropy density fabric channel — binding 9.
Thermo boundary	Maintains coherence with previous frame — prevMaintCost in receipts.
ThermoAccountant	Per-frame entropy ledger SSBO binding 2 — entropyThisFrame receipt.
Threat panel	https://127.0.0.1:9477/ — command, packets, threats, DNS, library, system.

Term	Definition
Tidewalker tier	Full spiderweb graph override — highest operator mirror control.
Time is linear	Axiom — logs are timeline; sealed clocks do not rewrite physics time.
TotalTime	ELLIE session clock API — seal and verify forward-only genesis.
Truth DNS	Loopback-first resolver — RFC trace, no foreign shortcut, field-dns.py.
USER_OK	Sovereign verify clean verdict — opposite of SQUIDGIE.
Verify at receive	Receiver double-check ethic — DNS hash, DHCP conflict, time SQUIDGIE.
VGA	Text mode buffer at guest 0xB8000 inside Field Die.
Vulkan fabric	rtx() singleton — device, queues, hardwareFabric, VRAM budget.
WASM surrogates	Replacement layer for dead plugins — preserves surface without JVM/Flash VM.
WebRTC	Realtime peer API held in Queen — gatekeeper per peer, not disabled.
Widevine	DRM system held via EME — license flows scored, not amputated.
x86.comp	Field Die compute shader — guest interpretation and Big Grin HUD.

Term	Definition
Zachary Geurts	Architect and author — AMOURANTHRTX, NEXUS-Shield, Queen, Field Primer.
ZMM1024	Tile cache tail in FieldX86Die SSBO — shader-side fabric sample cache for HUD.

[← Return to Field Technology v5 home](#)

Evidence anchor — grep and sources

Major claims in this chapter anchored for reproducibility. **Implemented** = grep today;
Metaphor = intuition; **Philosophy** = discipline.

Claim	Statement	Label	Evidence
Glossary	Canonical term readings	Implemented	This chapter
Master rocks	Consolidated honesty	Implemented	#master-rocks appendix
Disambiguation	Overload splits	Implemented	Phi vs RF vs entropy layers
Linear read	Required?	Philosophy	Grep table – don't read A–Z linearly

```
term_status ∈ {Implemented, Metaphor, Philosophy, Visual}
```

Anchor: #master-rocks in this chapter

Source paths

- content/chapters/22.html
- scripts/master_rocks_table.py

Chapter summary — before you turn the page

Glossary indexed — overloaded words disambiguated. Keep beside any chapter; definitions formal, rocks still visible.